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FIRST PROGRESS REVIEW

Doctor of Philosophy (PhD) in Computer Science

Chapter 2

Literature Review

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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Abbreviations

Abbreviation	Meaning
AI	Artificial intelligence
AUC	Area under the curve
AUROC	Area under the receiver operating characteristic curve
BraTS	Brain Tumor Segmentation challenge

Abbreviation	Meaning
CI	Confidence interval
CONSORT-AI	Consolidated Standards of Reporting Trials, AI extension
CT	Computed tomography
DECIDE-AI	Developmental and Exploratory Clinical Investigations of Decision support systems driven by AI
DSC	Dice similarity coefficient
ECE	Expected calibration error
EHR	Electronic health record
FL	Federated learning
FLAIR	Fluid-attenuated inversion recovery
FedAvg	Federated averaging
GPU	Graphics processing unit
H&E	Haematoxylin and eosin
HD95	95th-percentile Hausdorff distance
ICC	Intraclass correlation coefficient
IDH	Isocitrate dehydrogenase
LLM	Large language model
LMIC	Low- and middle-income country
LMICs	Low- and middle-income countries
MC	Monte Carlo
MCP	Model Context Protocol
MDT	Multidisciplinary team
MIL	Multiple-instance learning
ML	Machine learning
MRI	Magnetic resonance imaging
NCCP	National Cancer Control Programme
NPV	Negative predictive value
OOD	Out of distribution
PPV	Positive predictive value
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QA	Quality assurance
SSL	Self-supervised learning
SUS	System Usability Scale
TCGA	The Cancer Genome Atlas
WHO	World Health Organization
WSI	Whole-slide image

2. Literature Review

This chapter reviews the literature on which the framework builds and identifies, for each objective, the point at which existing work stops. It updates the review in the accepted proposal with the Research Paper Bank assembled during the first months of candidature. The chapter first describes how the literature was searched, selected and recorded (Section 2.1), then reviews the clinical and data context (Section 2.2) and each objective in turn: multi-agent orchestration (O1, Section 2.3), uncertainty and deferral (O2, Section 2.4), data-efficient, multi-source and federated models (O3, Section 2.5) and clinical evaluation of human-AI teams (O4, Section 2.6). Section 2.7 draws out themes that cut across the objectives, Section 2.8 synthesises what is established and what remains open, and Section 2.9 gives evidence tables for each objective.

2.1 Review Method

2.1.1 Search and selection

The review is a structured narrative review organised by objective. Searches were run in PubMed and PubMed Central, arXiv, Google Scholar, IEEE Xplore and the ACM Digital Library, and the proceedings of MICCAI, NeurIPS, ICML, ICLR, CVPR and AAAI were browsed directly. Search strings combined a domain term (brain tumour, glioma, neuro-oncology, medical imaging, computational pathology, clinical decision support) with a method term for each objective: multi-agent, orchestrator, agentic, tool use (O1); calibration, uncertainty, selective prediction, learning to defer, conformal prediction, abstention, out-of-distribution (O2); self-supervised, transfer learning, missing modality, multiple-instance learning, foundation model, federated learning, differential privacy (O3); and human-AI collaboration, reader study, automation bias, workload, usability, trust, reporting guideline (O4). Reference lists of key papers were followed backwards and forwards.

Inclusion criteria were: (i) relevance to at least one objective; (ii) publication from 2021 onwards, except for seminal methods that define a field (for example temperature scaling, deep ensembles, selective classification and learning to defer); (iii) peer-reviewed venue or a preprint from an established group, with predatory journals excluded; (iv) an open-access full text, so that every paper can be read and re-checked; and (v) enough methodological detail to extract the study design, data and evaluation. As advised at the supervision

meeting of 5 November 2026, priority was given to papers that describe their methodology in detail, to studies with external validation, and to studies reporting negative results.

The collection grew in two rounds. By 9 November 2026, 314 papers had been collected and their methodology extracted. A second round on 10 November 2026 added 313 open-access papers and removed 68 that had no open-access full text, were duplicates, or no longer matched the objectives, giving the current bank of **559 papers** (Table 2.1). Of these, 524 (94%) were published in 2021 or later, 299 appeared in journals or conference proceedings and 260 are arXiv or medRxiv preprints; 13 were designated *base papers* whose design the framework adopts or extends directly.

Table 2.1: The Research Paper Bank by objective (as of 10 November 2026).

Objective	Theme	Papers	Base papers
O1	Multi-agent architecture, orchestration, uncertainty hand-off and audit trail	143	4
O2	Calibrated uncertainty and escalation (when to defer to the clinician)	109	3
O3	Data-efficient MRI and pathology models, fusion and federated learning	146	4
O4	Clinical evaluation: performance, escalation appropriateness, workload, usability, trust and safety	161	2
Total		559	13

Figure 2.1 summarises how the collection was assembled. Because the bank was built for a doctoral programme rather than a single systematic review question, it is not a PRISMA systematic review; the flow diagram is given so that the reader can see how many papers entered, how many were removed and why, and how the final set is divided between the objectives.

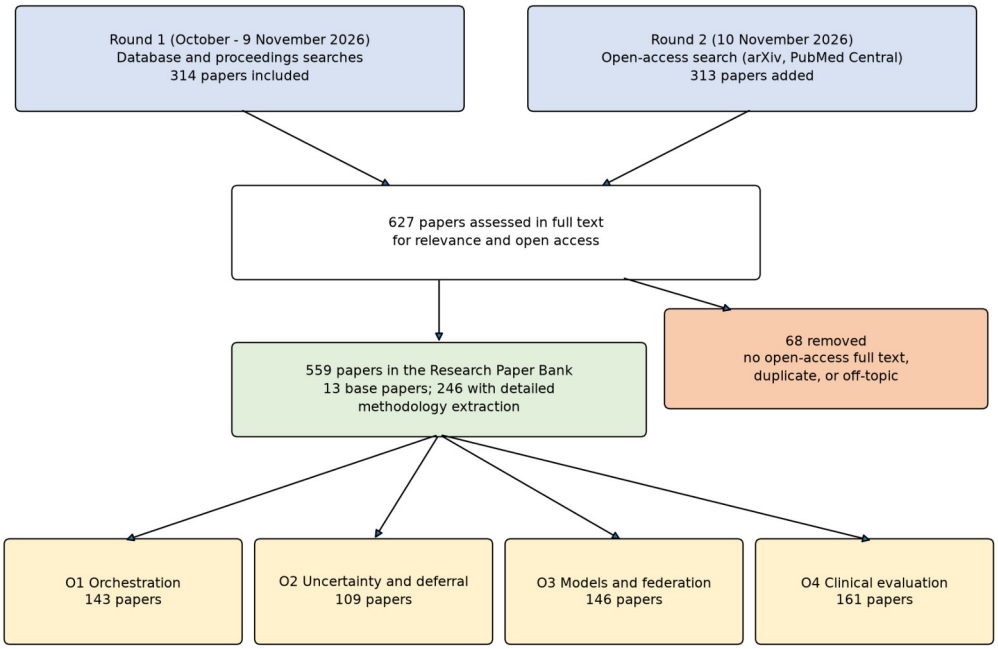


Figure 2.1: Flow of papers into the Research Paper Bank (two collection rounds, October-November 2026).

Table 2.2: Search terms used for each objective (combined with the domain terms brain tumour, glioma, neuro-oncology, medical imaging, computational pathology or clinical decision support).

Objective	Method terms
O1	multi-agent; orchestrator; agentic; LLM agent; tool use; ReAct; tumour board; clinical workflow; audit trail; accountability; model cards
O2	uncertainty quantification; calibration; temperature scaling; Monte Carlo dropout; deep ensemble; selective prediction; reject option; learning to defer; triage; conformal prediction; risk control; out-of-distribution; failure detection; abstention
O3	segmentation; nnU-Net; transformer; missing modality; self-supervised; transfer learning; semi-supervised; domain generalisation; foundation model; multiple-instance learning; whole-slide image; fusion; radiogenomics; federated learning; differential privacy; harmonisation
O4	reader study; human-AI collaboration; automation bias; reliance; explainability; workload; usability; acceptance; trust; reporting guideline; randomised trial; low- and middle-income countries; neurosurgery; tractography; surgical planning

2.1.2 Appraisal of the evidence

Each study was appraised for the features that matter most for this research rather than with a single scoring instrument. Five questions were asked: (i) was the evaluation internal only, or was there an external or prospective test; (ii) were the data clinical or benchmark data, and from which populations; (iii) was uncertainty or confidence reported, and was it calibrated; (iv) was a human expert part of the evaluation, and how; and (v) were code and data available. These questions recur in the evidence tables at the end of this chapter (Tables 2.6 to 2.9). Two features of the evidence base should be noted at the outset. First, 260 of the 559 papers are preprints, mostly in the fast-moving areas of language-model

agents and conformal prediction; their findings are treated as provisional and are cited for ideas and design rather than as established results. Second, very few studies come from South Asia, and none from Sri Lanka, which is itself a finding that motivates the local validation in Objective 3.

2.1.3 Data extraction

For every paper a structured record was completed in a review sheet per objective: aim, method, main result, tools, research gap, relation to this study and limitations. For every paper a detailed methodology extraction was then completed from the full text (559 of 559 papers, finished on 11 November 2026), covering study design and institutions, datasets and sample sizes, preprocessing, model architecture, training settings, baselines, evaluation metrics, statistical tests and availability of code and data. These records are published in the Research Paper Bank on the progress website, where each paper links to its source. The extraction directly informs the methodology in Chapter 3: baselines, metrics and statistical tests were chosen from what comparable studies used.

2.2 Clinical and Data Context

Brain and nervous-system tumours are uncommon but carry disproportionate mortality (Price et al., 2024). Sri Lanka sees roughly 700 new cases and 587 deaths a year (International Agency for Research on Cancer, 2024; National Cancer Control Programme, 2024). The most consequential recent change for computational work is the 2021 WHO classification, which integrates histology with molecular markers to define tumour type and grade (Louis et al., 2021). A model trained on a pre-2021 label is therefore not predicting the same target as one aligned to the current classification, and single-source systems can only support a diagnosis, never produce one of record. Surgery is constrained by the trade-off between extent of resection and preservation of eloquent function (Sanai & Berger, 2008; Molinaro et al., 2020).

The data on which most published models are built differ from the Sri Lankan setting. BraTS 2021 provided about 2,000 multi-institutional glioma cases, but mostly from Western centres and in pre-processed, co-registered form (Baid et al., 2021). BraTS-Africa was created to test whether such models work on routine-care MRI from Sub-Saharan Africa, acquired on older or low-field scanners with poorer contrast and resolution in patients who present later (Adewole et al., 2026); the winning team reached a lesion-wise Dice of 0.86 only after segmentation-aware augmentation and ensembling of three

architectures on the small local dataset (Ankomah et al., 2025). Low- and middle-income settings also face shortages of scanners, specialists and pathology services, delayed presentation and poor incidence data (Yevudza et al., 2024), and AI evaluations there are few and rarely real-world (Ciecierski-Holmes et al., 2022). These observations motivate the data-efficient, locally validated and federated approach of O3.

2.2.1 Tumour types, imaging and the integrated diagnosis

The tumours the registry will see most often are gliomas, meningiomas and brain metastases, with a smaller number of lymphomas, paediatric tumours and other entities. Gliomas are diagnosed under the 2021 WHO classification by combining histology with molecular markers, chiefly IDH mutation, 1p/19q co-deletion, ATRX loss and, for glioblastoma, features such as TERT promoter mutation; MGMT promoter methylation informs treatment (Louis et al., 2021). Standard pre-operative MRI includes T1-weighted, contrast-enhanced T1, T2-weighted and FLAIR sequences, often with diffusion and perfusion imaging and, for surgical planning, diffusion tensor imaging for tractography. In routine practice, sequences are frequently missing or acquired with different protocols, which is why robustness to missing sequences is a recurring theme below.

Imaging-only prediction of molecular status is an active field but not a substitute for tissue. A meta-analysis of 104 studies found pooled test-cohort sensitivity of 0.80 and specificity of 0.85 for IDH, and lower figures for 1p/19q (Farahani et al., 2026). A careful evaluation of MGMT promoter methylation prediction from MRI found all models close to chance (AUC 0.49-0.63) (Saeed et al., 2023), a negative result that cautions against claims of 'virtual biopsy'. Public datasets that link MRI to molecular markers, such as the UT Southwestern glioma dataset, are now appearing (Reddy et al., 2026), but the definitive diagnosis remains integrated and multi-source, which is why the framework places the diagnosis at a dedicated fusion agent with a clinician validating every diagnosis of record.

2.2.2 Public datasets available for development

Table 2.3 lists the public datasets most relevant to development and external testing. They allow every method to be built and benchmarked before local data are available, but all differ from Sri Lankan routine care in scanners, protocols, pre-processing and case mix (Chan et al., 2025).

Table 2.3: Public datasets relevant to this research.

Dataset	Content	Use in this study
BraTS 2021 (Baid et al., 2021)	About 2,000 multi-institutional glioma MRI cases (T1, T1c, T2, FLAIR) with expert segmentations; MGMT labels for a subset	Training and benchmarking of MRI segmentation (WP2)
BraTS-Africa (Adewole et al., 2026)	Routine-care glioma MRI from Sub-Saharan Africa, often on older or low-field scanners	External test under scanner and population shift
BraTS-PEDs (Kazerooni et al., 2024)	Paediatric brain-tumour MRI with segmentations	Robustness check for paediatric cases
BraTS-MEN-RT (LaBella et al., 2026)	750 meningioma radiotherapy-planning MRIs with gross tumour volume labels	Meningioma segmentation baseline
FeTS (Zenk et al., 2025; Linardos et al., 2025)	Glioma data partitioned by institution for federated benchmarking	Simulated federation (WP5)
UTSW glioma (Reddy et al., 2026)	MRI with molecular markers and segmentations	Imaging-marker experiments
TCGA glioma slides (Chen et al., 2022b; Wang et al., 2023)	Whole-slide H&E images with genomic data for adult gliomas	Pathology baselines and fusion (WP4)
CoRe-BT (Rivera et al., 2026)	Multimodal radiology-pathology-text benchmark for brain-tumour typing	Fusion benchmarking

2.2.3 Neuro-oncology care in low-resource settings

Evidence from comparable health systems describes the conditions the framework must work in. Neuro-oncology in Sub-Saharan Africa is limited by scarce scanners, neurosurgeons, neuropathology and radiotherapy, delayed presentation and poor incidence data (Yevudza et al., 2024). A systematic review of AI in low- and middle-income countries found only ten studies that met its criteria, half without details of algorithms or training data, and repeated problems of reliability, workflow fit and usability (Ciecierski-Holmes et al., 2022). Deployment studies in rural clinics found AI decision support misaligned with short consultations and heavy workloads (Wang et al., 2021a). Where AI has been evaluated carefully in such settings, results are encouraging: AI grading of diabetic retinopathy in a randomised trial raised referral adherence from 39.6% to 51.5% (Mathenge et al., 2022), and AI auto-segmentation reduced radiotherapy contouring from about an hour to ten minutes of review (Kibudde et al., 2024). Reviews of AI for diabetic retinopathy and tuberculosis in these countries point to reduced cost and workload but uneven reporting (Onyeze et al., 2025; Vijayan et al., 2026), and a responsible-AI framework for low-resource settings emphasises governance and local ownership (Alzyood, 2026). These studies show that benefit in low-resource care depends on fitting the workflow and on local validation, both of which this study builds in.

2.3 Multi-Agent Orchestration in Clinical AI (O1)

2.3.1 From single models to agents

Agentic AI combines a reasoning core with tools, memory and the ability to act over several steps. Foundational work showed that interleaving reasoning with tool calls improves reliability (Yao et al., 2023), and general frameworks such as AutoGen and MetaGPT made multi-agent conversation and role assignment easy to build (Wu et al., 2023; Hong et al., 2023). A recent architecture for healthcare agents describes four components (planning, action, reflection and memory) and proposes evaluation on recommendation accuracy, task execution, adaptation and explainability (Liu et al., 2025a).

2.3.2 Medical multi-agent systems

Medical multi-agent systems fall into three groups. The first simulates a multidisciplinary team with language-model agents. MDAgents lets a moderator agent rate case complexity and assign a single agent, a team or an integrated care team, achieving the best accuracy on seven of ten medical benchmarks (Kim et al., 2024); MedAgents and the Multi-Agent Conversation framework showed similar gains from structured discussion, the latter on 302 rare-disease cases (Tang et al., 2024; Chen et al., 2025a). The second orchestrates specialised tools. An oncology agent that called vision transformers, MedSAM, OncoKB and guideline retrieval raised decision accuracy on 20 multimodal cases from about 30% to 87% (Ferber et al., 2025); MedRAX and CXRAgent coordinate chest X-ray tools, CXRAgent adding an evidence validator that rates each tool claim as high, medium or low confidence (Fallahpour et al., 2025; Lou et al., 2026). The third is closest to this study: orchestrated clinical workflows with a defined pathway. An orchestrator with seven red-flag agents supported secondary-headache diagnosis in primary care (Wu et al., 2026b); an emergency triage system used a weighted confidence score below 0.7 to raise acuity, keeping the clinician as final decision-maker (Sharma et al., 2026); a multi-agent system automated BT-RADS scoring for glioma follow-up MRI with 76.0% agreement with an expert neuroradiologist, against 57.5% for the original clinical reports (Jabal et al., 2026); and TumorBoard combined radiology, neuropathology, molecular and guideline agents with a claim-evidence ledger and a safety governor that deferred 84.2% of unsafe cases under evidence deletion (Liu et al., 2026c).

At the most autonomous end, an agent operating in a simulated, FHIR-based electronic health record managed 574 emergency cases with 88.9% diagnostic accuracy and

outperformed board-certified physicians on matched cases (Ferber et al., 2026). Under clinical-scale workloads, orchestrated multi-agent designs sustained accuracy far better than a single agent handling all tasks (65.3% versus 16.6% at 80 concurrent tasks) and were cheaper to run (Klang et al., 2025).

2.3.3 Failure, consensus and accountability

The same literature documents how such systems fail. An analysis of 1,642 execution traces from seven frameworks found failure rates of 41% to 86.7%, caused by system design (44.2%), inter-agent misalignment (32.3%) and weak verification (23.5%) (Cemri et al., 2025). An audit of medical multi-agent systems over 14,400 cases found that unsupported observations entered 16.6% of cases and propagated downstream, that agents simply repeated their initial view in 98.4% of discussions, and that final answers showed authority bias in 28.8% (Zhu et al., 2025a). Optimising each component separately does not optimise the system: a best-of-breed system with excellent information-gathering scored lower diagnostic accuracy (67.7%) than an integrated multi-agent design (77.4%) (Bedi et al., 2025). Language models alone also remain unreliable for clinical decisions, misreading laboratory results and failing to follow guidelines (Hager et al., 2024). Governance work provides documentation and audit frameworks—model cards (Mitchell et al., 2019), internal algorithmic auditing (Raji et al., 2020) and the FUTURE-AI consensus (Lekadir et al., 2025)—but these are organisational processes rather than run-time mechanisms.

2.3.4 Foundations of language-model agents and collaboration

Several general mechanisms underpin current medical agents. Communicative agents with assigned roles (CAMEL) produced outputs preferred by human evaluators in 76% of comparisons (Li et al., 2023); multi-agent debate improved factual accuracy and reasoning over a single model, for example from 77.0% to 85.0% on grade-school mathematics (Du et al., 2023); and a mixture of open-source agents with an aggregator exceeded a frontier model on an instruction-following benchmark (Wang et al., 2024b). Surveys of language-model-based autonomous agents and multi-agent systems map these designs and identify the recurrent problems of hallucination, coordination and evaluation (Wang et al., 2024c; Guo et al., 2024a; Xi et al., 2025). A review of trust, risk and security management for agentic multi-agent systems, covering 180 primary studies, argues that governance,

explainability, privacy and model operations must be designed in from the start (Raza et al., 2026).

2.3.5 Benchmarks and simulated clinical environments

Because real clinical deployment is rare, agents are mostly evaluated in simulations. Agent Hospital simulated an entire hospital in which doctor agents improved by treating simulated patients (Li et al., 2024b). AgentClinic showed that models that score highly on static examination questions lose much of their accuracy when information must be gathered sequentially from a simulated patient (Schmidgall et al., 2024), and AI Hospital and ClinicalLab reached similar conclusions for multi-department diagnosis (Fan et al., 2024; Yan et al., 2025). In a virtual electronic health record, the best model completed only 69.7% of clinical tasks, with action tasks harder than queries (Jiang et al., 2025). MedAgentBoard compared multi-agent collaboration with conventional methods across medical tasks and found that collaboration does not always help (Zhu et al., 2025b). An evaluation of language models on 2,400 real abdominal cases concluded that they were not ready for autonomous decisions, misreading laboratory values and ignoring guidelines (Hager et al., 2024). For neurology, a multi-agent system with retrieval reached 94.6% accuracy on clinical-reasoning questions, while retrieval alone reduced the accuracy of one medical model from 52.9% to 41.2% (Sorka et al., 2025), showing that design choices can harm as well as help.

2.3.6 Imaging and pathology agents

Agents that coordinate imaging tools are most relevant to the MRI and Pathology Agents of this framework. MMedAgent learned to select medical tools across modalities (Li et al., 2024a); MedAgent-Pro followed an evidence-based workflow that raised balanced accuracy for glaucoma from 56.4% (GPT-4o alone) to 90.4% (Wang et al., 2025b); and VILA-M3 combined a vision-language model with expert models to outperform Med-Gemini (Nath et al., 2025). For brain imaging specifically, an agentic language model performed training-free neuroradiological image analysis by calling tools (Erdur et al., 2026), and LLM assistance raised junior radiologists' AUC for pre-operative glioma classification from 0.77 to 0.83 (Li et al., 2025a). In a usability study of LLM-assisted brain MRI differential diagnosis, assisted readers were correct in 61.4% of cases against 46.5% unassisted (Kim et al., 2025b). In pathology, multi-agent systems analyse whole-slide images with task allocation and verification (Lyu et al., 2025), a multi-agent copilot

supported evidence-based reasoning in human pathology (Weishaupt et al., 2025), and a language model with retrieval diagnosed neuropathological tumour subtypes correctly in 90% of tested cases, but not without retrieval (Hewitt et al., 2024). A review of agentic systems in computational pathology concluded that most are retrospective prototypes at low autonomy levels with no proven clinical benefit and no regulatory clearance (Lu et al., 2026).

2.3.7 Tumour boards and neuro-oncology decision support

Tumour boards are the clinical analogue of a multi-agent system: specialists contribute evidence and a decision emerges. Language models have been compared with breast, sarcoma, melanoma and molecular tumour boards with mixed agreement (Sorin et al., 2023; Dehdab et al., 2026; Babic et al., 2026; Schmutz et al., 2025), and in neuro-oncology radiotherapy a prospective comparison found that the model omitted indicated treatments in straightforward cases and suggested unnecessary therapy in borderline ones (Tini et al., 2025). A review of language models as decision aids in neuro-oncology highlights shared decision-making but cautions about reliability (Lawson McLean et al., 2024). Multi-agent designs for oncology add structure: EvoMDT outperformed single models on cancer question benchmarks (Liu et al., 2026b), a neuro-symbolic multi-agent system for trial matching reached F1 0.82 against 0.47 for GPT-4 zero-shot (Loaiza-Bonilla et al., 2026), and the Healthcare Agent Orchestrator summarised patients for molecular tumour boards (Blondeel et al., 2025). A thoracic tumour-board system was developed, evaluated and deployed, with an LLM judge agreeing with physicians in 89% of ratings (Ellis-Caleo et al., 2026). Closest to this study, TumorBoard addresses longitudinal neuro-oncology with a claim-evidence ledger and a safety governor (Liu et al., 2026c), and a multi-agent system automated BT-RADS scoring for glioma follow-up (Jabal et al., 2026). None of these, however, attaches calibrated uncertainty from imaging and pathology models to the board's reasoning.

2.3.8 Uncertainty, abstention and routing in agents

A growing body of work asks how language-model agents can know when they are unsure. Semantic entropy, which measures uncertainty over meanings rather than word sequences, predicted correctness better than token-level entropy (Kuhn et al., 2023) and detected hallucinations across 30 task-model combinations with AUROC 0.79 (Farquhar et al., 2024). Black-box uncertainty methods for closed models (Lin et al., 2024b), conformal

factuality guarantees that raised correctness from about 30% to 80% by removing unsupported claims (Mohri & Hashimoto, 2024), and surveys of calibration in language models (Liu et al., 2025b) provide tools, but a position paper argues that the aleatoric-epistemic split does not fit interactive agents (Kirchhof et al., 2025), and a review of uncertainty in medical language models proposes a broader framework (Atf et al., 2025). KnowNo used conformal prediction to make a robot planner ask for help only when needed, cutting human help by 10-24% at a guaranteed success rate (Ren et al., 2023), an idea transferable to clinical escalation.

In medicine, uncertainty-aware abstention improved selective performance on clinical texts (Vazhentsev et al., 2026), selective prediction for extracting data from clinical notes abstained on 43% of notes and cut total cost by 51% (Swaminathan et al., 2023), and a review of abstention in clinical language models proposed a decision-theoretic framework with abstention operating characteristics (Presacan et al., 2026). Confidence-aware collaboration between doctor and pharmacist agents improved diagnosis and medication accuracy by about 7 percentage points (Zheng et al., 2025), and confidence estimation has been proposed for trustworthy agents working on electronic health records (Song et al., 2025). Two recent preprints bring calibrated uncertainty into agent routing: Monte Carlo dropout uncertainty helped a decision-support agent only when communicated as a binary error-risk flag, cutting confident misdiagnoses from 8.5% to 2.7% (Hauke et al., 2026); and CRC-Router applied conformal risk control to routing on chest radiographs (Li et al., 2026), as did an uncertainty-aware agentic triage system for chest X-rays (Li et al., 2025b). An explainable agentic framework for stroke imaging with an abstention agent shows the same direction for neuroimaging (Islam, 2026). Learning-to-defer methods that consider which human expert to ask (Mozannar et al., 2023a; Hemmer et al., 2023) and benchmarks for deferral with realistic synthetic experts (Alves et al., 2025) provide the theoretical basis.

2.3.9 Governance, monitoring and audit

Accountability for a composite decision requires records of what each component did. Model cards document intended use and evaluation conditions (Mitchell et al., 2019), internal algorithmic auditing defines an end-to-end audit process (Raji et al., 2020), and FUTURE-AI sets 30 consensus recommendations including traceability (Lekadir et al., 2025). For deployed imaging AI, ReclAIm used a multi-agent system to monitor performance and trigger fine-tuning when decline was detected, restoring performance to

within 2% of baseline after drops of up to 40.6% (Tzanis & Klontzas, 2025), and hospital 'AI quality improvement' units have been proposed to monitor performance and distribution shift (Feng et al., 2022). Safety-focused architectures such as Polaris, a constellation of specialist models with safety checks, performed on par with nurses on safety dimensions (Mukherjee et al., 2024), and MedGuards uses agents to detect and correct medical errors (Ma et al., 2026). Orchestration of explainability for multimodal, longitudinal imaging is still immature: of 15 explainability libraries, only one supported joint multimodal input (Pahud de Mortanges et al., 2024). These works show the components of an auditable system but not an architecture in which audit is part of every hand-off.

2.3.10 Gap for O1

Existing medical multi-agent systems route by question difficulty, role or language-model self-assessment; confidence, where present, is qualitative or heuristic (Lou et al., 2026; Sharma et al., 2026). Only a handful of recent preprints attach statistically grounded uncertainty to routing (Li et al., 2026; Hauke et al., 2026), and they address a single routing step on chest radiographs. No published architecture for the brain-tumour pathway makes calibrated uncertainty part of the contract between agents, propagates it across dependent hand-offs, and records every AI output and clinician validation in an auditable trail. This is the contribution addressed by O1 and RQ1.

Table 2.4: Representative multi-agent clinical systems and what they lack relative to this study.

Study	Domain and design	Main result	Missing for this study
Kim et al. (2024)	LLM agents; moderator assigns team size by complexity	Best on 7/10 medical QA benchmarks	No calibrated confidence; no clinician in the loop
Ferber et al. (2025)	Oncology; GPT-4 calling vision and retrieval tools	Decision accuracy ~30% to ~87% (20 cases)	No deferral rule; tool errors pass silently
Lou et al. (2026)	Chest X-ray; director with evidence validator	67.0% CheXbench; preferred by radiologists	Confidence is High/Medium/Low, not calibrated
Sharma et al. (2026)	Emergency triage; orchestrator, four agents	80% exact ESI match, 0% undertriage (60 vignettes)	Heuristic weights; simulated cases only
Jabal et al. (2026)	Glioma follow-up MRI; extractor and scorer agents	76.0% vs 57.5% BT-RADS agreement (492 exams)	No uncertainty or escalation
Liu et al. (2026c)	Neuro-oncology tumour board; evidence ledger, safety governor	Deferred 84.2% of unsafe cases	Preprint; synthetic benchmark; no imaging models
Li et al. (2026)	Chest X-ray routing with conformal risk control	Best risk-coverage trade-off	Single step; no dependent chain or clinician cost

2.4 Uncertainty, Calibration and Deferral (O2)

2.4.1 Estimating uncertainty

The premise of the framework is that an agent must know when it does not know. Monte Carlo dropout approximates Bayesian inference in deep networks (Gal & Ghahramani, 2016) and deep ensembles provide a simple, strong alternative (Lakshminarayanan et al., 2017); separating aleatoric from epistemic uncertainty clarifies which part more data can reduce (Hüllermeier & Waegeman, 2021). Reviews of uncertainty quantification in medical deep learning group methods into Bayesian, ensemble, test-time augmentation, deterministic single-pass and post-hoc families (Lambert et al., 2024b; Loftus et al., 2022). In brain-tumour segmentation, the QU-BraTS challenge ranked uncertainty methods and found ensembles and test-time augmentation common among the best teams (Mehta et al., 2022), segment-level uncertainty tracked Dice and could flag poor segmentations (Mehrtash et al., 2020), and conformal prediction produced voxel-wise uncertainty that correlated strongly with Dice ($r = -0.83$) on internal and external glioma data (Elyassirad et al., 2025). For IDH prediction from MRI, a confidence-informed network trained on 2,481 gliomas from eight institutions reached 96% accuracy with a Brier score of 0.013 (Bangalore Yogananda et al., 2025). In pathology, uncertainty-thresholded predictions raised external AUROC from 0.93 to 0.99 (Dolezal et al., 2022), and an uncertainty-aware ensemble separated glioblastoma from lymphoma and flagged 67 unseen tumour types with AUROC 0.92 where Monte Carlo dropout reached only 0.67 (Zhao et al., 2025).

2.4.2 Calibration and its fragility

Modern networks are over-confident, and temperature scaling corrects most of this on in-distribution data (Guo et al., 2017). The central negative result is that calibration degrades under dataset shift, exactly where it is most needed; temperature scaling fitted on clean validation data does not hold, and deep ensembles are the most robust option (Ovadia et al., 2019). Medical benchmarks confirm the problem: no confidence-scoring function reliably prevented silent failures in medical image classification, and none consistently beat the plain maximum softmax response (Bungert et al., 2023; Bernhardt et al., 2022; Jaeger et al., 2023). In pathology, ensembles gave the best uncertainty under domain shift and label noise (Mehrtens et al., 2023b). Multi-domain temperature scaling, which uses heterogeneity across sites, improves robustness to shift (Yu et al., 2022), a result directly relevant to a federation of a few heterogeneous hospitals.

2.4.3 Acting on uncertainty: selective prediction, deferral and conformal guarantees

Selective classification gives a model a reject option and guarantees the error on accepted cases: on ImageNet, 2% top-5 error was guaranteed at about 60% coverage (Geifman & El-Yaniv, 2017). Calibrated selective classification further requires the accepted cases to be well calibrated, including under a new hospital (Fisch et al., 2022). Learning to defer goes further by modelling the human expert: the system learns a predictor and a rejector jointly with a consistent surrogate loss (Mozannar & Sontag, 2020; Madras et al., 2018), later made calibrated with one-vs-all classifiers (Verma & Nalisnick, 2022) and extended to multiple experts, cost-sensitivity and workload constraints (Verma et al., 2023; Alves et al., 2024). Triage formulations allocate scarce human effort explicitly (Raghu et al., 2019; Okati et al., 2021), and sequential deferral has been studied in simulated treatment settings (Joshi et al., 2022). In medicine, combining deferral with uncertainty reached the same F1 while deferring far fewer patients (for example 36% versus 69% for pleural effusion) (Liu et al., 2022), and complementarity-driven deferral to clinicians reduced false positives and clinician reads in real screening programmes (Dvijotham et al., 2023).

Conformal prediction provides distribution-free coverage guarantees for any model (Angelopoulos & Bates, 2021), and conformal risk control extends them to arbitrary monotone losses such as the false-negative rate (Angelopoulos et al., 2024a). Medical applications have flagged unreliable predictions in prostate pathology (one error in 794 predictions versus fourteen without conformal prediction) (Olsson et al., 2022), reduced false positives of a head-CT model at a new hospital from 45% to 5% while abstaining on 14% of examinations (Angelopoulos et al., 2024b), and combined conformal selection with cost-aware deferral for sepsis triage under shift (Kwon & Kim, 2026). In brain-tumour MRI classification, withholding 1.8% of predictions cut selective risk by a third (Alshamlan, 2026). The guarantees, however, are marginal rather than per patient, assume exchangeable data, and can fail for subgroups and under shift (Mehrtens et al., 2023a); federated conformal prediction is only beginning to address distributed sites (Lu et al., 2023).

2.4.4 Measuring calibration and the limits of calibration metrics

Calibration is usually summarised by expected calibration error, but the measure has known weaknesses. Calibration tests for multi-class problems provide a unifying statistical framework (Widmann et al., 2019), and information-theoretic analysis shows how the

binning in expected calibration error affects its bias (Futami & Fujisawa, 2024). Revisiting calibration across modern architectures found that vision transformers and MLP-Mixers are among the best calibrated and degrade less under shift (Minderer et al., 2021), and foundation models change the picture further: post-hoc calibration helps them in distribution but can become counterproductive under severe shift (Hekler et al., 2025). Importantly, calibration and failure detection are different goals: most calibration methods were useless or harmful for predicting individual failures (Zhu et al., 2022), and in medical image classification self-supervised pre-training improved calibration (Sambyal et al., 2023). For this study, calibration error will therefore be reported together with failure-detection measures, not as a proxy for them.

2.4.5 Single-pass and large-model uncertainty

Ensembles and Monte Carlo dropout require many forward passes, which matters for 3D MRI. Deep deterministic uncertainty matched five-member ensembles on out-of-distribution detection with a single pass (Mukhoti et al., 2023), and large pre-trained models with reliability extensions improved uncertainty across tasks as model size increased (Tran et al., 2022). Evidential deep learning gives uncertainty in one pass; a region-based evidential method for brain-tumour segmentation achieved the best uncertainty-error overlap among evidential variants (Li et al., 2022). Out-of-distribution detection is surveyed in general (Yang et al., 2024) and for medical imaging specifically (Hong et al., 2024), and the OpenMIBOOD benchmark provides medical out-of-distribution test sets (Gutbrod et al., 2025). These options let the framework trade computational cost against uncertainty quality for each agent.

2.4.6 Uncertainty and failure detection in brain-tumour imaging

Brain-tumour studies show both the value and the limits of uncertainty. Exploiting uncertainty in a segmentation network raised enhancing-tumour Dice by 3.15 points (Lee et al., 2022); uncertainty maps for post-operative glioblastoma segmentation were judged useful, with 72% of segmentations rated acceptable by three physicians (Holtzman Gazit et al., 2022); and uncertainty-aware segmentation of meningioma radiotherapy targets improved Dice and Hausdorff distance while adding a map that a standard network lacks (Wang et al., 2025a). In radiogenomics, the lowest-uncertainty EGFR predictions were 83% accurate against 48% for the most uncertain (Hu et al., 2021a), and a calibrated craniopharyngioma classifier revealed that an earlier published model had been over-fitted

(Prince et al., 2023). Uncertainty estimates for segmentation can be unreliable at the subject level (Jungo & Reyes, 2019), and how voxel uncertainty is aggregated to a case-level score matters: spatially aware aggregation outperformed simple averaging (Guarino et al., 2026), soft Dice confidence was near-optimal for selective segmentation including on BraTS (Borges et al., 2026), and pairwise Dice between ensemble members was a strong and simple failure detector across segmentation tasks (Zenk et al., 2024). Quality-control frameworks that flag failed segmentations raised mean Dice on accepted cases from 0.82 to 0.84 (Williams et al., 2021), and counterfactual auditing of glioma segmentations detected failures across MRI datasets (Le et al., 2026). A task-aware evaluation of uncertainty for glioma diagnosis found that a composite trust score did not consistently beat classification uncertainty for selective prediction (Rojas et al., 2026).

2.4.7 Learning to defer: theory and extensions

Learning to defer has developed rapidly since the first consistent estimator (Mozannar & Sontag, 2020). Joint learning of predictor and rejector is more sample-efficient than staged learning when expert labels are scarce (Charusaie et al., 2022), and good deferral is possible with only a few expert predictions per class—six per class reached 99% of the full-label accuracy in a radiology task (Hemmer et al., 2023). Softmax parametrisation can be made both calibrated and consistent (Cao et al., 2023), realizable consistent losses give stronger guarantees (Mao et al., 2024b), and fine-tuning improves deferral (Raman & Yee, 2021). Extensions handle multiple experts and populations of experts (Mao et al., 2024a; Tailor et al., 2024; Keswani et al., 2021), top-k experts (Montreuil et al., 2025), multiple users who may be complemented or deferred to (Zhang et al., 2024b), coverage constraints (Zhang et al., 2024c), and multiple objectives through post-processing (Charusaie & Samadi, 2024). Combining human labels with model probabilities through confusion matrices roughly halved error relative to either alone on image benchmarks (Kerrigan et al., 2021). Asymmetric abstention, which rejects differently for the two error types, improved biomedical decisions without rejecting more cases (Gandouz et al., 2021)—the closest existing idea to the cost-asymmetric threshold of this study.

2.4.8 Conformal prediction in medical imaging

Conformal methods are increasingly applied to medical images. Learn-then-test calibrates algorithms to control risks such as false-discovery rate (Angelopoulos et al., 2025), and selective conformal risk control combines selection with risk guarantees (Xu et al., 2025).

Group-conditional conformal prediction reduced coverage gaps between skin types (Lu et al., 2022), reliable-region conformal prediction met very low error targets where classical methods failed (Zhang et al., 2023b), and conformal prediction under ambiguous ground truth showed that voted labels can under-cover expert disagreement (Stutz et al., 2023). For segmentation, conformal methods give pixel-wise sets (Mossina et al., 2024), control false positives (Mossina & Friedrich, 2025), respect anatomy (Gaillochet et al., 2026), give asymmetric volumetric risk (Alijani et al., 2026) and produce robust volume estimates under covariate shift (Lambert et al., 2024a; Cheung et al., 2025); in prostate MRI, conformal prediction improved volume agreement with the reference (Gade et al., 2024). Reviews of clinical conformal prediction note that studies are isolated and rarely involve practising clinicians (Vazquez & Facelli, 2022; Angelopoulos & Bates, 2023). In pathology, uncertainty estimates detected 70-90% of mispredictions (Poceviciute et al., 2022), and benchmarks under domain shift favour ensembles (Mehrtens et al., 2023b; Fayjie et al., 2024).

2.4.9 Deferral that reduces workload in practice

Workload reduction is the practical promise of deferral. In mammography, AI identification of normal examinations could exclude 19.1% of examinations without missing a cancer (Lang et al., 2021), and entropy-guided review of automatic sleep scoring improved agreement with the gold standard while sending about 60% of epochs to review (Kang et al., 2021). Self-supervised learning with abstention improved diabetic retinopathy screening safety (Chopra et al., 2026), and a benchmarking study of Bayesian deep learning for diabetic retinopathy showed that uncertainty-based referral improves accuracy on retained cases but degrades under shift (Band et al., 2022; Filos et al., 2019). These results define the operating points that a clinical escalation policy should aim for, and the shift problem that it must monitor.

2.4.10 Gap for O2

Three limitations define the gap. These formulations address one model making one decision, not a chain in which uncertainty must be composed across agents. They assume a fixed, homogeneous cost of error, whereas clinical costs are asymmetric and differ by decision type. And they model the expert as always available, whereas specialist time is the binding constraint (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024). Large-language-model agents add a further difficulty: the usual aleatoric/epistemic

split does not transfer to interactive agents (Kirchhof et al., 2025), and how uncertainty is *communicated* to a downstream agent changes whether it is used (Hauke et al., 2026). A calibrated escalation policy over a dependent chain, with clinically elicited asymmetric costs and finite capacity, is the contribution of O2 and RQ2.

2.5 Data-Efficient, Multi-Source and Federated Models (O3)

2.5.1 MRI segmentation and classification

U-Net-based architectures remain the backbone of brain-tumour segmentation. nnU-Net configures itself to each dataset and matched or beat specialised methods on 23 public datasets (Isensee et al., 2021); a later re-evaluation found that many claimed improvements over it disappear under rigorous validation (Isensee et al., 2024). Transformer designs such as Swin UNETR and TransBTS (Hatamizadeh et al., 2022; Wang et al., 2021b) and scaled convolutional networks such as MedNeXt (Roy et al., 2023) are competitive, and promptable foundation models (MedSAM, MedSAM2) generalise across modalities but need prompts and struggle with small or unclear boundaries (Ma et al., 2024; Ma et al., 2025). On real-world clinical data with missing sequences, nnU-Net remained within or above the range of human inter-rater agreement (Dice 0.86 internal, 0.93 external) (Pemberton et al., 2023). Routine scans often lack sequences; dedicated models learn shared representations that tolerate missing modalities, with the largest gains when few sequences are available (Zhang et al., 2022b; Liu et al., 2023).

2.5.2 Learning with few labels

Because annotated local data will be scarce, label efficiency is essential. Self-supervised pre-training of Swin transformers on unlabelled 3D volumes improved downstream segmentation after fine-tuning (Tang et al., 2022), and stepwise transfer learning achieved expert-level paediatric tumour segmentation from limited data, with AI contours judged acceptable more often than expert contours (80.2% versus 65.4%) (Boyd et al., 2024). Even limited target-domain adaptation yields rapid gains in cross-dataset brain-tumour segmentation (Grande et al., 2026b). What remains unestablished is whether representations learned this way produce confidence estimates fit to drive an autonomy decision; most evaluations report accuracy only.

2.5.3 Computational pathology and fusion

Attention-based multiple-instance learning lets a single slide-level label supervise attention over thousands of tiles (Ilse et al., 2018), and CLAM made this data-efficient and interpretable (Lu et al., 2021). Pathology foundation models such as UNI now serve as general-purpose encoders with strong few-shot transfer (Chen et al., 2024c), although clinical benchmarks show that their performance varies by task and site (Campanella et al., 2025) and that they do not remove site-specific batch effects (Kömen et al., 2024). In brain tumours, deep learning reached neuropathologist-level integrated classification of adult diffuse gliomas (AUC above 0.90 for all major types) (Wang et al., 2023), and intra-operative molecular classification has been achieved with stimulated Raman histology and ultra-fast sequencing, the latter abstaining rather than making confident errors (Hollon et al., 2023; Vermeulen et al., 2023). Fusion of histology with genomic features outperformed either source and WHO 2016 grading (Chen et al., 2022b), and a mixture-of-experts fusion of pathology and radiology foundation models reached AUC 0.94 on an independent test set for WHO 2021 subtyping while still working when only one modality was present (Saueressig et al., 2026). Multimodal radiology-pathology-text benchmarks are beginning to appear (Rivera et al., 2026).

2.5.4 Federated and privacy-preserving learning

Federated averaging trains a shared model by exchanging parameters rather than data (McMahan et al., 2017); FedProx and SCAFFOLD correct for heterogeneous clients (Li et al., 2020; Karimireddy et al., 2020), and FedBN keeps batch normalisation local to absorb scanner differences (Li et al., 2021b). Early medical studies showed feasibility without sharing patient data (Sheller et al., 2020), and the decisive demonstration trained a glioblastoma boundary model across 71 sites on six continents (Pati et al., 2022). The FeTS challenges benchmarked aggregation methods and exposed worst-case sites hidden by good average Dice (Zenk et al., 2025; Linardos et al., 2025). An extensive benchmark found that plain FedAvg already performs very well for brain-tumour segmentation, with other methods giving only small gains (Manthe et al., 2024). An international paediatric platform lost less than 1.5% in classification and 3% in segmentation relative to centralised training (Lee et al., 2024a), and federated evaluation platforms such as MedPerf allow benchmarking at sites without moving data (Karargyris et al., 2023). Personalised methods address appearance shift and modality mismatch between hospitals (Jiang et al., 2022;

Wang et al., 2022; Dai et al., 2024), and differentially private federated learning has been shown to be viable for medical images (Adnan et al., 2022; Abadi et al., 2016).

2.5.5 Architectures and foundation models for brain MRI

Beyond nnU-Net and Swin UNETR, transformer hybrids such as TransBTS, NestedFormer and CKD-TransBTS model inter-sequence relations explicitly (Wang et al., 2021b; Xing et al., 2022; Lin et al., 2023), and MedNeXt scales convolutional networks with transformer-inspired design (Roy et al., 2023). Foundation models are reaching neuroimaging: BrainSegFounder, pre-trained on large unlabelled neuroimaging collections, beat previous challenge winners on BraTS and stroke-lesion data, with larger models and more pre-training data helping (Cox et al., 2024); MedSAM2 extended promptable segmentation to 3D and cut annotation cost by more than 85% in large annotation studies (Ma et al., 2025); and adapters make the general Segment Anything model usable for medical images (Wu et al., 2025), although an experimental study found its zero-shot medical performance highly variable (Mazurowski et al., 2023). Foundation models for glioma genotyping from multi-parametric MRI are also emerging (Farahani et al., 2025a; Farahani et al., 2025b). For the registry, promptable models are attractive as annotation assistants, with clinicians correcting rather than drawing contours.

2.5.6 Missing sequences, metastases and real-world data

Real data rarely include every sequence. Adversarial co-training kept Dice usable when only one or two sequences were present, where earlier methods fell to 10-20% (Wang et al., 2021d); latent correlation learning, mmFormer, M3AE and alignment-based methods train a single model for all combinations (Zhou et al., 2021; Zhang et al., 2022b; Liu et al., 2023; Liu et al., 2024a). In a multicentre study of brain metastases, a model trained to handle missing sequences performed as well as one requiring all sequences (Grovik et al., 2021), and nnU-Net remained robust to missing sequences on real-world glioma data (Pemberton et al., 2023). Metastasis detection reached 95.8% sensitivity with 0.39 false positives per patient and raised radiologists' sensitivity (Yin et al., 2022); in a randomised crossover multi-reader study, AI assistance raised contour Dice from 0.87 to 0.92 and shortened contouring time (Luo et al., 2024); and 2.5D and 3D models on multinational data showed the trade-off between sensitivity and false positives across centres (Ottesen et al., 2022). For paediatric tumours, a single-centre pathologically diverse dataset gave Dice 0.85 for whole tumour, close to human variability, but only 0.26 for cystic components

(Piffer et al., 2026). These studies show what clinically useful performance looks like and where it fails.

2.5.7 Semi-supervised learning and domain adaptation

Semi-supervised learning uses unlabelled cases directly: uncertainty-guided mutual consistency selects reliable predictions for consistency training (Zhang et al., 2021), a teacher-student approach addressed annotation scarcity in brain-tumour segmentation with the student surpassing the teacher on subregions (Liu et al., 2026a), and semi-supervised contrastive learning improved label efficiency (Hu et al., 2021b). Self-supervised methods for 3D MRI include 3D SimCLR with Monte Carlo dropout (Ali et al., 2021), imbalance-aware self-supervised radiomics (Li et al., 2021a), contrastive few-shot tumour detection (Chin et al., 2024) and a self-supervised vision transformer for tumour classification (Karagoz et al., 2024). Domain shift is addressed by unsupervised cross-modality adaptation (He et al., 2021), causality-inspired single-source generalisation (Ouyang et al., 2021), generalisation across heterogeneous sequences (Zhang et al., 2025; Grande et al., 2026b), self-training with tumour-aware deformations (Grande et al., 2026a) and annotation-efficient target-modality adaptation (d'Assier et al., 2022). Active learning that combines transfer learning with selective annotation can reduce labelling for tumour classification (Hao et al., 2021). These methods are the toolkit for WP2, where local labels will be few and local scanners different.

2.5.8 Computational pathology for brain tumours

Multiple-instance learning dominates slide-level analysis (Gadermayr & Tschuchnig, 2022): TransMIL models correlations between tiles and reached AUC 0.93 on CAMELYON16 against 0.88 for CLAM (Shao et al., 2021), CAMIL adds tissue context (Fourkioti et al., 2023), and hierarchical self-supervised vision transformers scale to gigapixel images (Chen et al., 2022a; Chen & Krishnan, 2022). Self-supervised pre-training reaches baseline performance with 40% of the labels (Schirris et al., 2021), generalises biomarker prediction across cohorts (Niehues et al., 2023; Saldanha et al., 2023) and produces phenotype representations (Ye et al., 2024; Hou et al., 2024), while lightweight self-supervised models offer domain generalisation with lower computing cost (Shafique et al., 2025). Vision-language pathology foundation models reach strong zero-shot subtyping (Lu et al., 2024); benchmarks compare adaptation strategies (Lee et al., 2024b), and a neuropathology-specific foundation model captured brain-tissue features

better than general models (Verma et al., 2025). Batch effects survive in foundation-model features and are not removed by stain normalisation (Kömen et al., 2024), which motivates stain augmentation methods such as RandStainNA (Shen et al., 2022).

For gliomas specifically, a multi-centre assessment reached 0.79 accuracy internally and 0.73 on an external test set for five WHO CNS5 types (Jin et al., 2023), transfer learning classified diffuse glioma subtypes with balanced accuracy 0.87 (Im et al., 2021), a single-centre weakly supervised model inferred IDH status with AUC 0.95 on TCGA (Zuo et al., 2025), deep learning on pathology images stratified survival risk (Chunduru et al., 2022; Luo et al., 2023), low-rank contrastive learning improved grading on both FFPE and frozen sections (Zhang et al., 2022a), and integrating molecular data with pathology improved classification (Pei et al., 2021). Review work sets out the challenges of segmentation and classification in glioma pathology (Kurc et al., 2020), and a subtype study combined in-domain and out-of-domain transfer learning (Despotovic et al., 2023). For paediatric tumours, UNI with attention-based MIL was best but performance fell from in-site to out-of-site testing (Tampu et al., 2026). Intra-operative work includes frozen-section discrimination of lymphoma from glioma that outperformed several pathologists (Zhang et al., 2024a), label-free optical imaging for molecular classification (Hollon et al., 2023), ultra-fast sequencing-based classification during surgery that abstained rather than misclassified (Vermeulen et al., 2023), and methylation classification from sparse data (Brändl et al., 2025) concordant with final neuropathology in 89% of cases (Djirackor et al., 2021).

2.5.9 Radiology-pathology fusion

Fusion of imaging and histology predates the 2021 classification: a pipeline combining radiology and histology data reached balanced accuracy 0.86 for glioma classification (Hamidinekoo et al., 2020), and weakly supervised fusion of whole-slide images and multi-parametric MRI ranked third in the CPM-RadPath challenge (Hsu et al., 2022). Pathomic Fusion combined histology with genomics (Chen et al., 2022b), and the latest mixture-of-experts fusion of foundation-model features aligned to WHO 2021 subtypes still worked when only one modality was present (Saueressig et al., 2026). The CoRe-BT benchmark now supports robust multimodal typing with radiology, pathology and text (Rivera et al., 2026). The gap is that these studies assume curated, complete cohorts and report no calibrated confidence for the fused decision.

2.5.10 Federated learning: heterogeneity, pathology and privacy

Data heterogeneity is the main technical obstacle in medical federation. FedDG learns domain-general features by sharing amplitude spectra and improved Dice on unseen sites (Liu et al., 2021); personalised methods adapt the shared model to each site (Jiang et al., 2023; Wang et al., 2024a; Wang et al., 2022; Lin et al., 2024a); and federated evidential active learning selects which local cases to annotate under domain shift (Chen et al., 2024a). Federation with missing modalities has been addressed with modality-specific encoders (Dai et al., 2024). For brain tumours, early feasibility work showed multi-institutional training without data sharing (Sheller et al., 2018), an international paediatric platform lost little accuracy relative to central training (Lee et al., 2024a), federated IDH prediction from MRI was feasible but incurred a cost relative to central training (Bas & Ozturk-Isik, 2026), federated radiomics generalised to an external dataset of 466 patients (Ren et al., 2025), and Byzantine-robust aggregation protected heterogeneous brain-MRI federation from corrupted updates (Karami et al., 2026). Federated learning was also feasible for multiple-sclerosis lesion segmentation across real sites despite heterogeneity (Hindawi et al., 2025). Studies that apply federation to brain-tumour classification on public MRI collections (Albalawi et al., 2024; Islam et al., 2022) are less informative than the multi-institutional studies above, because their sites are simulated.

Computational pathology has its own federated literature: federated learning on gigapixel slides (Lu et al., 2020), swarm learning across three centres (Saldanha et al., 2022), nationwide deployment behind hospital firewalls in Germany (Babendererde et al., 2026), dropout-based and cross-assessment aggregation for histopathology (Gunesli et al., 2021; Zhang et al., 2023a), and a systematic review of federated histopathology classification (Touhami et al., 2026). Privacy threats in federated medical learning are real—gradients can leak information (Das et al., 2024)—so differential privacy (Adnan et al., 2022; Singh et al., 2020; Nampalle et al., 2023), privacy-preserving harmonisation (Galada et al., 2024), few-shot synthetic data generation for sharing (Shi & Wang, 2025) and broader privacy reviews (Zhu et al., 2024) are relevant. Surveys and benchmarks of federated medical image analysis summarise methods and show that results depend strongly on how data are split (Guan et al., 2023; Zhou et al., 2025), and federated foundation models are beginning to be explored (Liu et al., 2024b). Finally, the intersection of federated learning and uncertainty quantification is identified as a gap by two reviews (Koutsoubis et al., 2024;

Koutsoubis et al., 2025), with federated conformal prediction an early answer (Lu et al., 2023).

2.5.11 Gap for O3

Three gaps remain. Most fusion methods assume complete, high-quality inputs, whereas in practice histology lags imaging, slide quality varies and molecular markers may be unaffordable. Label-efficient and federated methods are validated almost entirely on Western, curated cohorts and large, well-resourced sites; the regime of a handful of small, heterogeneous hospitals is little characterised. And federation is studied as a training-time issue in isolation: how aggregation changes calibration, and therefore the behaviour of a deferral policy validated on an earlier model, is essentially unexamined (Koutsoubis et al., 2024; Koutsoubis et al., 2025). These define O3 and RQ3.

2.6 Clinical Evaluation of Human-AI Teams (O4)

2.6.1 Reporting and validation standards

Reporting guidelines now cover each stage of clinical AI evaluation: DECIDE-AI for early-stage live evaluation of decision support, with explicit items on human factors and safety (Vasey et al., 2022); CONSORT-AI for randomised trials (Liu et al., 2020); TRIPOD+AI for prediction models (Collins et al., 2024); and STARD-AI for diagnostic accuracy studies (Sounderajah et al., 2025). Metrics Reloaded recommends combining overlap and boundary metrics for segmentation and calibration metrics when probabilities are used (Maier-Hein et al., 2024). A scoping review of 86 randomised trials of AI found mostly single-centre studies with diagnostic-yield endpoints and poor demographic reporting (Han et al., 2024).

2.6.2 Does the team beat its parts?

The evidence is mixed and instructive. On average, human-AI combinations underperformed the better of human or AI alone in decision tasks (Malone et al., 2024). The effect of AI assistance varied greatly between radiologists, could not be predicted from experience, and incorrect AI predictions harmed performance (Yu et al., 2024); clinicians were susceptible to inaccurate advice whatever its claimed source (Gaube et al., 2021). Explanations increased reliance on wrong advice (Bansal et al., 2021) and had variable effects across clinicians (Nicolson et al., 2025). Positive results come where the human can

weigh the AI's reliability case by case: endoscopists followed AI more when it was right than when it was wrong, and the team beat both (Reverberi et al., 2022). Cognitive forcing functions reduced over-reliance at a cost in perceived usability (Bucinca et al., 2021). Most relevant to this study, selective prediction reduced the overall harm of automation bias but changed the error pattern: when told the AI was abstaining, clinicians missed more diagnoses and treatments (Jabbour et al., 2025). Automation bias also appears under time pressure in computational pathology (Rosbach et al., 2024) and in neuroradiology reading (Kim et al., 2025a).

2.6.3 Workload, usability and trust

A meta-analysis found that concurrent AI assistance reduced reading time by 27%, and AI pre-screening reduced reading volume by 62%, with sensitivity maintained or improved (Chen et al., 2024b). In mammography, automatic identification of normal examinations cut workload by 17% while excluding 1% of true positives (Rodriguez-Ruiz et al., 2019), and AI-integrated screening pathways improved single-reader performance (Frazer et al., 2024). In a low- and middle-income radiotherapy setting, AI auto-segmentation reduced contouring from about an hour to ten minutes of human review (Kibudde et al., 2024). Usability and trust are measured with established instruments—the System Usability Scale, NASA Task Load Index and Likert trust scales (Brooke, 1996; Hart & Staveland, 1988)—and trust is shaped by clinical experience, perceived risk and how uncertainty is presented (Cao et al., 2024).

2.6.4 Surgical planning and neurosurgery

AI in neurosurgery is reviewed broadly (Awuah et al., 2024; Baker et al., 2024), and machine-learning-based trajectory planning to the tumour has been demonstrated (Dundar et al., 2022). Navigated tractography is accurate for motor and language pathways in glioma surgery (Ille et al., 2021), and automated post-operative segmentation can standardise extent-of-resection assessment (Luque et al., 2024; Zanier et al., 2023). These works evaluate geometric accuracy rather than surgeon acceptability, and none presents the uncertainty of the underlying anatomy alongside a recommended route.

2.6.5 Reliance, explanation and decision-making

The human-AI decision literature shows that appropriate reliance—following the AI when it is right and not when it is wrong—is the goal, not trust as such. A decision-theoretic

framework separates loss due to mis-reliance from loss due to poor information (Guo et al., 2024b), and measurement constructs for appropriate reliance are being standardised (Raees & Papangelis, 2026). People overestimate their own competence in ways that reduce appropriate reliance (He et al., 2023); learning from feedback improves it (Schemmer et al., 2023); onboarding with learned rules raised team accuracy by 5.2% (Mozannar et al., 2023b); and time pressure changes the accuracy-time trade-off and reliance (Swaroop et al., 2023). Explanations rarely enable complementary performance because most tasks do not allow the explanation to be verified (Fok & Weld, 2023), proxy tasks and subjective measures can mislead when evaluating explainable AI (Buçinca et al., 2020), and interface design shapes collaboration in high-stakes decisions (Chen et al., 2025b). Presenting AI uncertainty affects reliance, moderated by the user's initial decision and background (Cao et al., 2024); adjusting for trust can reduce inappropriate reliance (Srinivasan & Thomason, 2025); and majority voting among doctors improved the appropriateness of reliance in pathology (Gu et al., 2024). These findings shape the clinician interface in Objective 1: the system shows calibrated confidence and the reason for escalation, and the study measures appropriate reliance rather than raw agreement.

2.6.6 Automation bias in imaging and pathology

Automation bias—over-reliance on automated advice—is well documented. AI assistance shortened reading time for cerebral aneurysm detection at all experience levels but exposed readers to automation bias (Kim et al., 2025a); in computational pathology, time pressure increased the severity of automation bias (Rosbach et al., 2024), and anchoring on suggestions persisted across pathologists (Rosbach et al., 2026). In mammography, automation bias degraded performance in multi-reader settings but improved it for single readers (Frazer et al., 2024), and reader bias changed with cancer prevalence and AI support (Al-Bazzaz et al., 2024). Explainable AI did not remove automation bias (Schemmer et al., 2022), and reviews describe deskilling and cognitive risks of generative AI (Al-Anezi, 2026; Bittencourt, 2025). The most relevant finding is that selective prediction reduced the overall harm of automation bias but increased missed diagnoses and treatments when clinicians were told that the AI had abstained (Jabbour et al., 2025). An escalation system must therefore be evaluated on how clinicians behave on escalated cases, not only on how accurate the AI is on the others.

2.6.7 Reader studies in pathology and radiology

Reader studies give the design template for WP7. In pathology, AI assistance improved Gleason grading so that assisted pathologists outperformed both unassisted pathologists and the AI alone (Bulten et al., 2021); a liver-cancer assistant improved accuracy when correct but reduced it when wrong (Kiani et al., 2020); AI improved efficiency in prostate biopsies (Eloy et al., 2023), PD-L1 scoring (Wang et al., 2021c) and mitotic counting reproducibility (Bertram et al., 2022; Graham et al., 2026); and the mode of assistance mattered, with one mode robust to false-positive AI predictions (Chuang et al., 2026). In radiology, AI support improved fracture detection most for non-obvious fractures (Bachmann et al., 2024), improved diagnostic confidence and sensitivity especially for junior clinicians in emergency imaging (Novak et al., 2024), gave modest AUROC gains and shorter interpretation in breast ultrasound (Kwon et al., 2026), and improved early lung-cancer detection on radiographs (Dissez et al., 2022). Dual suggestions helped some specialties more than others (Wu et al., 2026a). In neuro-oncology, AI predictions raised physicians' glioma-grading accuracy from 82.5% to 87.7%, with explanations adding no further significant gain (Jin et al., 2024), and LLM assistance improved radiologists' brain-tumour diagnosis in a constrained setting (Song et al., 2026). A large public dataset of radiologist-AI interactions now supports secondary analysis (Moehring et al., 2025).

2.6.8 Workload, triage and safety evidence

Beyond the meta-analysis of workload reduction (Chen et al., 2024b), triage studies quantify potential savings: AI as an additional reader suggested a 69.5% workload reduction in a hybrid triage scenario (Seker et al., 2024), simulated breast-screening triage reduced workload but missed some microcalcification cancers (Riccardi et al., 2026), and a narrative review summarises workload effects across specialties (Jeong et al., 2025). Safety evidence is less developed: a systematic review linked AI to patient-safety outcomes (Choudhury & Asan, 2020), an analysis of FDA device reports found that a quarter of reports were potentially AI-related while a third lacked enough information to tell (Handley et al., 2024), and reviews describe AI's potential for safety in general and in neurosurgery (De Micco et al., 2024; Sugiyama et al., 2025). Randomised trials of AI remain few and mostly single-centre (Han et al., 2024; Lam et al., 2022; Park et al., 2022); examples include AI-assisted colonoscopy (Xu et al., 2021) and deterioration analytics (Keim-Malpass et al., 2026). A systematic review of COVID-19 imaging models found

none fit for clinical use, mainly because of bias and poor validation (Roberts et al., 2021), a warning for any rapidly developed model.

2.6.9 Usability, acceptance and trust

Clinician acceptance follows technology-acceptance models: in a meta-analysis, intention to use AI decision support correlated with performance expectancy (r = 0.66), effort expectancy, social influence and facilitating conditions (Dingel et al., 2024); reviews of barriers and facilitators, national surveys and cross-disciplinary studies confirm these drivers (Scipion et al., 2025; Shi et al., 2026; Breitwieser et al., 2025; Fujimori et al., 2022), and human and technological factors interact (Knop et al., 2022). Usability methods for decision support include heuristic evaluation, live usability testing and dual methods (Cho et al., 2022; Richardson et al., 2019; Rizvi et al., 2025), with design lessons such as preferring brief, flexible, non-interruptive support (Blecker et al., 2019) and user-centred design (Chyzhyk et al., 2025; Blanes-Selva et al., 2023; Kunz et al., 2025); reviews summarise usability assessment and explainable decision support (Cubukcu & Canbaz, 2025; Abbas et al., 2025). Trust depends on clinicians' experience and perceived risk (Schulz et al., 2025), on explanations that can increase or decrease it (Rosenbacke et al., 2024), on workload and perceived risk (Shamszare & Choudhury, 2023), and on contextual explanation (Davis & Salwei, 2026); reviews of trust in clinical AI (Asan et al., 2020; Jones et al., 2023) and studies of trust calibration (Sakamoto et al., 2024; Lin-Martore et al., 2025; Cicin & Çetin Gürkan, 2026) show that the aim is calibrated trust. Prospective implementations show that early experiences shape sustained adoption (Naicker et al., 2026) and that user satisfaction can be good even when outcomes do not change significantly (Smith et al., 2025). Neurosurgical teams are broadly positive about AI but want it for imaging interpretation, planning and hazard alerts rather than autonomous surgery (Layard Horsfall et al., 2021; Boaro et al., 2025).

2.6.10 Pre-operative planning: tractography, extent of resection and visualisation

The surgical-planning agent builds on established neurosurgical tools. Tractography of the corticospinal tract and language pathways is accurate for subcortical resection (Ille et al., 2021), navigated tractography identifies high-risk language areas (Salvati et al., 2021), combining navigated transcranial magnetic stimulation with tractography helps preserve motor function, with greater distance to the tract associated with stable function (Eibl et al., 2024), and constrained spherical deconvolution and multi-level fibre tracking improve tract

reconstruction in glioma (Sheng et al., 2021; Zhylka et al., 2021). A preoperative tractography algorithm has been proposed for tumours in the motor pathway (Ordonez-Rubiano et al., 2023), and connectome-guided surgery is reviewed as an emerging paradigm (Patel et al., 2026). Automated extent-of-resection assessment on early post-operative MRI was reliable and prognostic (Luque et al., 2024), although another model determined extent of resection correctly in only 44.1% of cases (Zanier et al., 2023), and editing AI contours affected gross tumour volume accuracy (Hochreuter et al., 2024). Virtual and augmented reality support planning and awake craniotomy (Mofatteh et al., 2022; Gómez Amarillo et al., 2023; Zawy Alsofy et al., 2021), with open-source virtual-reality frameworks and markerless augmented reality under development (Fiederer et al., 2019; Ho et al., 2024; Hellum et al., 2022; Asadi et al., 2024), and intra-operative imaging reviews look towards AI integration (Bin-Alamer et al., 2023). Registration error estimation for MRI-ultrasound (Salari et al., 2023) and simulation for planning (Zhao et al., 2024) show how uncertainty can be made visible to surgeons. Reinforcement-learning path planning to the tumour has also been demonstrated (Dundar et al., 2022). None of these combines segmentation uncertainty with route planning, which is the specific feature the proof of concept will test.

2.6.11 Gap for O4

No reporting standard defines how to measure the quality of an escalation decision itself—whether the system deferred on the cases that needed a clinician and acted on the ones that did not—and evidence on how clinicians respond to a deferring system in brain-tumour care is absent. Evaluations in low-resource settings are rare. A multi-reader, multi-case evaluation that measures escalation appropriateness, missed escalations, workload, usability and trust in situ is the contribution of O4 and RQ4.

2.7 Cross-Cutting Themes

Reading across the four objectives, five themes recur and directly shape the design of this study.

1. **Evaluation is moving from outputs to processes.** Audits of multi-agent systems (Cemri et al., 2025; Zhu et al., 2025a), reliance frameworks (Guo et al., 2024b) and reporting guidelines (Vasey et al., 2022) all look at how a decision was reached, not only whether it was right. The audit trail of Objective 1 and the escalation-quality measures of Objective 4 follow this direction.

2. **Calibration is necessary but fragile.** Every objective depends on calibrated confidence, but calibration degrades under shift (Ovadia et al., 2019), after model updates and potentially after federated aggregation (Koutsoubis et al., 2024). Calibration is therefore treated as a quantity to monitor, not a property to assume.
3. **Simple baselines are hard to beat.** Maximum softmax response for failure detection (Jaeger et al., 2023; Bungert et al., 2023), nnU-Net for segmentation (Isensee et al., 2024) and FedAvg for federation (Manthe et al., 2024) repeatedly match more complex methods. Every experiment in this study includes these baselines.
4. **Human-AI teams can be worse than their parts.** Benefit depends on who decides which cases the AI handles (Dvijotham et al., 2023; Malone et al., 2024; Jabbour et al., 2025). This is the central argument for a principled escalation policy rather than AI assistance on every case.
5. **Local validation is the exception.** Few studies come from low- and middle-income settings (Ciecierski-Holmes et al., 2022), and none from Sri Lanka. A locally validated, federated and calibrated framework would add evidence where it is scarcest.

2.8 Synthesis

Four things are established. The perception tasks the framework requires are technically solved to a clinically useful standard on curated data, for MRI (Isensee et al., 2021; Pemberton et al., 2023) and for digitised slides (Lu et al., 2021; Wang et al., 2023). Tools for quantifying and calibrating uncertainty are mature (Guo et al., 2017; Lakshminarayanan et al., 2017; Angelopoulos & Bates, 2021), with formal mechanisms for acting on it (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020). Federated learning is a proven route to multi-institutional training without data sharing (Pati et al., 2022; Lee et al., 2024a). And both ethical guidance and regulation require demonstrable human oversight of high-risk clinical AI (World Health Organization, 2021; European Union, 2024).

Four things remain open, and together they define this research (Table 2.5). The primitives for selective autonomy have not been assembled into a calibrated escalation policy for a dependent chain of clinical sub-decisions with asymmetric costs and finite expert capacity. Agentic architectures do not treat uncertainty as part of the contract between agents, so hand-offs are unqualified and accountability for composite decisions is undefined. Multi-source diagnosis is demonstrated on complete, high-quality cohorts rather than under the

incompleteness of real practice. And data-efficient, federated and calibrated learning have been studied separately, leaving unexamined how federated updates and domain adaptation change calibration and thereby silently change the safety behaviour of an escalation policy.

Table 2.5: Summary of open problems and how this research addresses them.

Open problem	Key evidence	Addressed by
Escalation policy for a dependent chain with asymmetric costs and finite clinician time	(Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Alves et al., 2024; Dvijotham et al., 2023)	O2 / RQ2 (WP3)
Uncertainty as part of the contract between agents; auditable hand-offs	(Cemri et al., 2025; Zhu et al., 2025a; Li et al., 2026; Liu et al., 2026c)	O1 / RQ1 (WP3, WP1)
Fusion and perception robust to missing, poor or local data	(Chen et al., 2022b; Saueressig et al., 2026; Adewole et al., 2026; Boyd et al., 2024)	O3 / RQ3 (WP2, WP4)
Effect of federation and adaptation on calibration and deferral	(Manthe et al., 2024; Ovadia et al., 2019; Koutsoubis et al., 2024)	O3 / RQ3 (WP5)
Measuring escalation quality, workload and trust in situ	(Vasey et al., 2022; Malone et al., 2024; Jabbour et al., 2025)	O4 / RQ4 (WP6, WP7)

2.9 Evidence Tables

Tables 2.6 to 2.9 summarise representative studies for each objective from the structured extraction records of the Research Paper Bank: the design and data, the main result, and the main limitation in relation to this study. Base papers are included in every table. The full records for all 559 papers, with links to the sources, are available in the Paper Bank on the progress website.

Table 2.6: Evidence table for multi-agent orchestration (O1): 30 representative studies.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Kim et al. (2024) (base)	Retrospective benchmark (in-silico) evaluation of a training-free LLM multi-agent framework on public medical QA datasets.	Ten public benchmarks (50 random test samples each in main runs; 100 in Table 12 runs). Text: MedQA (USMLE, English test set 1,273 questions, 5 options)	GPT-4(V), 50 samples: MDAgents MedQA 88.7 +/- 4.0 vs best solo CoT-SC 83.9 +/- 2.7 and ReConcile 81.3 +/- 3.0; PubMedQA 75.0 +/- 1.0 vs ReConcile 79.7 +/- 3.2 (not best); Path-VQA 65.3 vs 62.1 (weighted voting)	Only question-answer benchmarks. No real patient pathway, no calibrated confidence, and the human doctor is not part of the loop.
Ferber et al. (2025) (base)	Proof-of-concept development and evaluation study using fictional (simulated) patient cases built from public and in-house images; retrospective use of public images (TCGA, TCIA) and images from University Hospital Aachen, Germany.	(1) Knowledge base: about 6,800 medical documents and clinical scores from six sources: MDCalc (clinical scores), UpToDate and MEDITRON guidelines.	Tool use 97% (32/33; one failed call for patient Z, one omitted call for patient Le); about 3 tool calls per case on average (1 for cases T and Z, 8 for case X).	Only 20 cases, not prospective, no deferral rule. Errors in one tool can flow silently to the final answer.
Dvijotham et al. (2023) (base)	Retrospective, multi-dataset simulation study of human-AI deferral.	(1) UK Mammography: CoDoC tuning data = 60,755 cases (1,198 biopsy-confirmed cancers) from the diagnostic AI's hyperparameter-tuning subset.	Table 1 (sensitivity/specificity %, CoDoC vs clinician vs AI). UK, defer to first reader OP1: CoDoC 72.6/91.8, reader 62.7/92.8, AI 64.9/93.9; OP2: 64.4/96.5. UK, defer to double read OP3: CoDoC 71.0/95.1, double read 67.6/96.1.	Deferral uses only the scalar AI confidence score; it does not co-train the diagnostic model and deferral model, does not use image features or case metadata.
Liu et al. (2025a) (base)	Narrative literature review and conceptual framework paper.	No dataset used.	No new quantitative results.	No quantitative synthesis or systematic appraisal of evidence. Explainability has no standard metric. Long-term learning of agents is hard to evaluate.
Liu et al. (2026c)	Method development with a 360-case hidden benchmark built by clinical authors; perturbation, ablation and expert-rating studies (Kunsan National University, preprint).	Longitudinal neuro-oncology case packets (serial MRI, pathology, molecular markers, treatment history, performance status, guideline versions)	Action F1 0.772 (95% CI 0.754-0.790), entailment 0.914, harmful release 0.039 vs 0.068 for the typed council; gain over best baseline 0.031 (Holm p=0.0012); governor reduced harmful release by 7.8 points at 4.3 points false deferral; 14,220 tokens and 21.8 s per case.	Deferral based on evidence sufficiency rules rather than calibrated model uncertainty; no imaging models; synthetic benchmark.
Chen et al. (2025a)	Retrospective, simulation-based evaluation study using published case reports; no human participants (ethics approval waived).	(1) Orphanet rare-disease database: pool of over 7000 rare diseases in 33 categories; 302 diseases sampled across all 33 categories.	Primary consultation (most likely / possible / helpful tests): GPT-3.5 = 16.23% / 27.92% / 47.68%; GPT-4 = 19.65% / 34.55% / 58.17%.	Uses text-only rare-disease case reports from the literature, not real prospective patients or images.
Islam (2026)	Conceptual framework with qualitative case-based evaluation (Daffodil International University, Bangladesh, preprint).	Representative acute ischaemic stroke CT/MRI slices (public data; no new cohort reported).	Abstention arose in diagnostically ambiguous regions and low-information slices; no new performance benchmark claimed.	Fixed thresholds, no calibration analysis or clinician evaluation; no quantitative results.
Wu et al. (2026b)	Retrospective, in-silico evaluation study on published case vignettes; no prospective patients.	Gold-standard set of 90 adult secondary-headache cases (free-text vignettes).	Precision / recall / F1. Qwen-30b: single QPrompt 0.498/0.806/0.542; single GPrompt 0.492/0.700/0.495; multi-agent QPrompt 0.559/0.717/0.563; multi-agent GPrompt 0.600/0.796/0.605 (best overall F1).	Best F1 of 0.605 is too low for clinical use. Tested only on curated published case reports, not real primary-care notes, and only 7 red flags.
Hauke et al. (2026)	Retrospective experimental study with a 2x2 factorial agent experiment (RWTH Aachen, Heidelberg, Dresden; preprint).	TAIX-Ray Thorax public cohort: 137,593 training, 34,860 validation, 42,928 test radiographs (patient-stratified); ordinal grades for eight findings (heart size, congestion, effusion, opacities, atelectasis left/right).	Uncertainty raised error-detection AUROC from 0.74 to 0.77 (+0.023, 95% CI +0.014 to +0.033); the agent used uncertainty only as a binary error-risk flag, cutting confident misdiagnoses on unreliable findings from 8.5% to 2.7%.	Chest X-ray only; agent simulated; no human clinicians.
Li et al. (2026)	Method development and retrospective evaluation, standalone and as a plug-in to an agent (University of Notre Dame, preprint).	NIH ChestX-ray14: 112,120 frontal CXRs, 14 labels; patient-level split 0.5 train / 0.25 calibration / 0.25 test, calibration split equally into risk-model training and CRC calibration.	Strongest risk-coverage trade-off among baselines, meeting the 5% risk target on more findings, both standalone and with MedRAX.	Uniform risk target per finding, no asymmetric clinical costs or clinician capacity; single step, not a chain of agents.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Erdur et al. (2026)	Method development and benchmark creation for agentic brain MRI analysis (TUM, Imperial; preprint).	Multicentre brain MRI cohort: pre-operative glioma (20), post-operative glioma (42), metastasis (24) and others, with longitudinal subsets (30 LUMIERE patients with expert RANO ratings)	Agentic LLMs could perform neuro-radiological 3D analysis without training, but accuracy fell with workflow complexity, especially longitudinal tasks; benchmark released.	No uncertainty from the segmentation tools is passed to the LLM; no clinician escalation.
Jabal et al. (2026)	Retrospective single-institution evaluation, Duke University Medical Center; IRB approved, consent waived.	509 consecutive post-treatment glioma MRI examinations; 492 evaluable (9 without suitable baseline, 8 segmentation QA failures); mean age 57.1; median interval 63 days.	Accuracy 374/492 (76.0%, 95% CI 72.1-79.6) vs 57.5% for initial clinical scoring (P<.001); BT-1b 100%, BT-1a 92.7%, BT-3a 87.5% sensitivity; threshold categories lower; BT-4 PPV 92.9%.	No uncertainty or escalation of borderline volume-threshold cases; single centre.
Ma et al. (2026)	Method development and benchmark evaluation (NYU Abu Dhabi, Khalifa University, preprint).	Medical error detection and correction benchmarks of clinical text (details in paper).	Improved detection and correction over single-model baselines.	Text errors only; no imaging; no calibrated confidence.
Wang et al. (2025b)	Retrospective benchmark study on public datasets plus clinician rating of outputs.	(1) REFUGE2: retinal fundus images for glaucoma diagnosis (binary); test size Not reported.	Glaucoma (REFUGE2) bAcc/F1: MedAgent-Pro 90.4/76.4; GPT-4o 56.4/21.1; Janus-Pro 53.4/13.3; LLaVA-Med 50.0/0.0; BioMedCLIP 58.1/21.3; Qwen2.5-VL 54.3/16.3; InternVL2.5 51.8/13.8; MedAgents 52.1/8.9; MMedAgent 52.4/16.3.	Depends on the availability of visual tools for each domain; qualitative steps still rely on a hallucination-prone VLM; the step reliability check is a VLM judgement.
Yan et al. (2025)	Retrospective benchmark and agent evaluation on real hospital records.	ClinicalBench: 1,500 real cases, 150 diseases x 10 cases each, 24 departments, average about 1,000 tokens per case, Chinese originals translated to English with GPT-4.	Single LLMs (Avg): InternLM2 51.98 (DG Acc 64.47, PD 78.20, CDR 31.40, Acceptability 11.11, DWR 14.91); Gemini-Pro 50.63; Yi-34B 50.53; GPT-4 49.45 (FD 54.20, CDR 33.27, Acceptability 11.37); Qwen-72B 48.92; HuatuoGPT2-34B 48.42.	Imaging is given only as text reports, not images. No uncertainty, calibration or deferral to clinicians. No few-shot or RAG tested.
Jiang et al. (2025)	Benchmark development and retrospective evaluation study using de-identified, time-jittered real patient data from Stanford Hospital (STARR), USA.	Patient profiles: 100 patients randomly sampled from a STARR cohort who had an inpatient sodium lab test ordered on the morning of 13 November 2023.	Overall SR / Query SR / Action SR: Claude 3.5 Sonnet v2 69.67% / 85.33% / 54.00%; GPT-4o 64.00% / 72.00% / 56.00%; DeepSeek-V3 62.67% / 70.67% / 54.67%; Gemini-1.5 Pro 62.00% / 52.67% / 71.33%	No agent is reliable enough for unsupervised clinical action, especially for write actions and multi-step tasks.
Fallahpour et al. (2025)	Retrospective technical development and benchmark evaluation study. University of Toronto, Vector Institute, University Health Network and Cohere, Toronto, Canada.	ChestAgentBench (new): 675 chest cases with CXRs and full documentation from Eurorad (European Society of Radiology peer-reviewed case database), CC BY-NC-SA 4.0.	ChestAgentBench accuracy (LLaVA-Med / CheXagent / Llama-3.2-90B / GPT-4o / MedRAX): Detection 32.4 / 38.7 / 58.1 / 58.7 / 64.1; Classification 30.8 / 34.7 / 56.5 / 54.6 / 62.9; Localization 30.2 / 42.5 / 59.9 / 59.0 / 63.6.	No uncertainty quantification, so the agent cannot say when it is unsure or hand over to a clinician. It struggles when tools disagree.
Lou et al. (2026)	Retrospective technical development and benchmark study with ablations plus a blinded pairwise reader study by five certified radiologists.	1) CheXbench (public): image perception and image-text reasoning tasks used: single disease identification (SDI) on MIMIC-CXR and CheXpert.	CheXbench overall accuracy: LLaVA-Med 36.0, CheXagent 61.8, Janus-Pro-7B 42.3, GPT-4o 59.9, Qwen-VL-Max 60.3, MedGemma 55.5, MedRAX (GPT) 57.1, MedRAX (Qwen) 56.7, CXRAgent (GPT) 66.3, CXRAgent (Qwen) 67.0.	Confidence from EDV is a qualitative LLM judgement (High/Medium/Low), not a calibrated probability, and there is no formal deferral rule to a clinician. Only chest X-ray.
Nath et al. (2025)	Retrospective model development and benchmarking study on public datasets. NVIDIA with SingHealth (Singapore) and NIH (USA). No prospective or clinical deployment.	Training mixture (Supp. Table 7, original count -> balancing factor -> balanced count): USMLE medical text 10,178 x10 = 101,780; RadVQA 6,281 x16 = 100,496.	Table 1 (VQA-RAD / MIMIC-VQA / SLAKE / PathVQA accuracy; BLEU-4; ROUGE; ChestX-ray14 F1; CheXpert F1; total average): Med-Gemini 78.8 / 78.6 / 84.8 / 83.3 / 20.5 / 28.3 / 46.7 / 48.3 / 55.7.	Brain MRI is only used through a 2D input plus a 3D BraTS expert; no quantitative brain-tumour segmentation or diagnosis results are reported for the VLM.
Ellis-Caleo et al. (2026)	Quality-improvement study: retrospective comparative evaluation (cases April-June 2025) followed by prospective deployment with post-deployment monitoring (from January 2026).	Randomly chosen 50 thoracic tumour-board cases from April to June 2025 (one academic quarter).	Judge validation: exact agreement 89.3-89.4%; Cohen kappa 0.687-0.709; Gwet AC1 0.870-0.871; Fleiss kappa 0.731 (95% CI 0.697-0.762) for 2 physicians and 0.708 (95% CI 0.684-0.734) with the LLM added.	Single centre, single cancer type, text notes only (no imaging or genomics attachments). No uncertainty or deferral mechanism; no 'single agent with all tools' baseline.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Raza et al. (2026)	Structured (systematic-style) literature review with bibliometric analysis. Vector Institute (Canada), Cornell University (USA), University of Groningen (Netherlands).	No patient data.	Screening: 250 shortlisted, 180 included. Corpus composition: journal articles 61.2%. Figure 1 cites more than 2,300 papers mentioning 'agentic AI' from January 2023 to April 2025. No quantitative system results.	No empirical validation of the framework or of CSS/TUE on a real system; weights in the composite scores are left to the user. Few clinical examples.
Vazhentsev et al. (2026)	Retrospective computational benchmark study. Authors from MBZUAI (Abu Dhabi, UAE), Sber AI Lab (Moscow, Russia) and partners.	1) MIMIC-III mortality (binary, discharge summaries, 2001-2012): 34,692 patients, 260,195 records, 2 classes, record length quartiles (4.9k, 8.4k, 11.6k), age (49, 64.	Table 1 (normalised RC-AUC first 50%; MIMIC MCP uses FR-AUC), as MIMIC mortality / OV / MIMIC MCP: SR 0.38+/-0.01 / 0.40 / 0.25+/-0.01; Entropy -0.27 / 0.37 / 0.17; Delta -0.27 / 0.38 / 0.14; Beta 0.37 / 0.40 / -0.07.	Only encoder-only classifiers; no generative decoder LLMs; no method guarantees high uncertainty on errors; hybrid methods need a labelled validation set.
Ferber et al. (2026)	Retrospective simulation study with head-to-head comparison against physicians (reader-style study).	MIMIC-IV v2.2 (about 300,000 patients).	Diagnosis vs MIMIC-IV (n = 574): mean 88.9%, appendicitis 98.6% (146/148), pancreatitis 92.3%, UTI 77.6%, pneumonia 72.4%.	Retrospective sandbox simulation only; patient speech derived from structured HPI text; single public dataset with possible training-data contamination.
Liu et al. (2026b)	System development plus multi-layer evaluation: (1) benchmark testing on six public medical QA datasets.	Public QA benchmarks (filtered for oncology relevance): MedQA (general medical exam MCQ); MedQA-Cancer (cancer-only subset of MedQA)	MCQ accuracy: EvoMDT MedQA 0.86, MedQA-Cancer 0.89, MedMCQA-Cancer 0.75; Llama3-70B 0.76/0.76/0.63; Claude-3 0.79/0.75/0.69. Generative BERTScore: EvoMDT MedQARo 0.66, MedQuAD 0.63, ROND 0.68; Claude-3 0.61/0.59/0.65.	Only four cancer types and retrospective Chinese data; no imaging or raw unstructured notes as direct input; no prospective deployment or patient outcome study.
Loaiza-Bonilla et al. (2026)	Prospective evaluation of consecutive patients, January to December 2024, at US academic-community sites (St.	Prospective cohort: 3804 consecutive patients with metastatic or progressive solid tumours, ECOG 0-2, balanced for cancer-type incidence; broad eligibility (active/progressing disease, candidates for further therapy).	Held-out test set: multi-agent system sensitivity 0.8375, specificity 0.8359, precision 0.8121, F1 0.8246 (95% CI about 0.81-0.85 in text; 0.81-0.83 in abstract).	Only identification efficiency measured, not consent or enrollment; single health system; commercial platform with author conflicts of interest; underlying models.
Lu et al. (2026)	Narrative review with an operational taxonomy; authors from Shanghai Eastern Hepatobiliary Surgery Hospital, Nanyang Technological University, Fudan University Huashan Hospital, National University of Singapore, PuzzleLogic Pte Ltd and UC San Diego.	No primary data.	Reported figures from reviewed studies: WSI-Agents 10-17% higher than WSI-LLaVA, MDAgents and Med-Agents; Renal-Agent mean quadratic-weighted kappa 0.73 over 16 Banff scores, balanced accuracy 0.97 for rejection subtype.	No prospective pathology cohort studies, few matched ablations separating agentic organisation from backbone and data, poor reporting of latency and cost.
Sharma et al. (2026)	Technical development and retrospective simulation-based evaluation on standardized published vignettes.	ESI Implementation Handbook v4 (AHRQ, 2012) case studies, text vignettes, N = 60: 30 practice cases and 30 competency cases.	Phase 1 combined: exact 66.67% [55.00, 78.33]; within +/-1 95.00% [88.33, 100]; priority accuracy 86.67% [76.67, 95.00]; sensitivity 87.18% [76.86, 97.44]; specificity 85.71% [66.67, 100].	Only simulated vignettes were used.
Presacan et al. (2026)	Systematic literature review (PRISMA) plus a theoretical decision-theoretic formalisation and a proof-of-concept simulation pilot; authors from Romania.	Review corpus: searches in PubMed, ACM DL, IEEE Xplore, Semantic Scholar plus manual Google Scholar/Google search; last search 23 September 2025.	Review: 28 included studies.	Few benchmarks test abstention in realistic, multi-turn clinical settings; most treat abstention as binary; no unified predictor joins epistemic and normative signals.
Dehdab et al. (2026)	Retrospective single-centre study at the Sarcoma Center, University Hospital Tübingen; IRB-approved, consent waived.	200 cases randomly drawn from 1,901 sarcoma tumour-board cases in 2024; after exclusions 152 soft tissue sarcoma cases, 13 used for prompt development and 138 analysed.	Scores significantly below the maximum in all five domains (P<0.0001); contextualisation best; no difference across sarcoma subtypes (H=19.74, P=0.138); errors mainly in sequencing and chemotherapy choice.	No confidence estimate or abstention; single LLM; compared with board decisions rather than outcomes.
Babic et al. (2026)	Retrospective single-centre cohort study, University Clinical Center Tuzla, Bosnia and Herzegovina (2020-2024); a resource-limited setting.	151 consecutive treatment-naive adults with histologically confirmed cutaneous melanoma discussed at the melanoma MDT; demographic, histopathological, BRAF, SLNB and AJCC stage variables extracted.	Selected LLMs showed useful concordance with the MDT; differences were largest in complex cases needing adjuvant or systemic treatment decisions.	No confidence or abstention; pragmatic chat use without logging of retrieved evidence; no prospective effect on decisions.

Table 2.7: Evidence table for uncertainty, calibration and deferral (O2): 30 representative studies.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Angelopoulos & Bates (2021) (base)	Tutorial / methods guide with worked examples and Python pseudocode. No new clinical study.	Worked examples on ImageNet classification (prediction sets), regression with quantile regression, and other examples (tumour segmentation, time series, outliers) in later sections.	Theorem: with exchangeable data, split conformal gives coverage of at least 1 - alpha for any model and any score. The usefulness (set size and conditional coverage) depends on the score function.	Guarantee needs exchangeable data, which is not true across hospitals.
Verma & Nalisnick (2022) (base)	Method paper with theory (consistency proof) and experiments on simulated and real expert data.	CIFAR-10 with a simulated expert (correct with probability 0.70-0.75 on some classes, random on the others)	CIFAR-10 ECE (%) OvA / softmax / proxy: random expert 0.68 / 3.72 / 2.83; random data 2.05 / 2.07 / 39.06; both useful 1.68 / 3.32 / 37.15. Main simulation ECE 3.01% (OvA) vs 7.58% (softmax).	Single expert; simulated experts on most datasets; does not handle a budget on expert time or different error costs.
Kwon & Kim (2026) (base)	Retrospective development with temporally separated in-distribution and out-of-distribution test splits.	Early sepsis prediction EHR data with temporally separated ID and OOD test splits (details in paper).	Retained-case error reduced 49.6% (ID) and 46.7% (OOD) at 80% coverage; expected cost low on both splits, moderate increase under shift; low ECE; near-perfect NPV at 95% sensitivity; coverage close to 90% ID, slightly lower OOD; weighted conformal most robust; gender coverage gap 1.4 pp with Mondrian.	Tabular EHR sepsis task, not imaging; single deferral threshold; costs assumed.
Rojas et al. (2026)	Retrospective task-aware evaluation of uncertainty quantification (Erasmus MC, Amsterdam UMC, GE HealthCare; MELBA 2026).	Pre-operative glioma MRI (T1w, T1wCE, T2w, FLAIR) from multi-centre public and institutional cohorts used for multi-task segmentation, IDH, 1p/19q and grade (details in paper).	Task-aware uncertainty behaved differently across tasks; the composite trust score did not consistently outperform classification uncertainty for selective prediction.	Single UQ family; no clinician deferral or cost analysis.
Li et al. (2025b)	Method development with five-fold evaluation on a public subset (University of Notre Dame, preprint).	Class-balanced NIH ChestX-ray14 subset of 1,000 frontal radiographs (500 pulmonary oedema positive, 500 negative), publicly available on Kaggle.	Both routers outperformed VLMs and supervised classifiers in full-coverage accuracy and AURC; rule-based and LLM routers gave complementary operating points (throughput vs accuracy).	Fixed heuristic thresholds without risk guarantees; small balanced dataset; one finding.
Chopra et al. (2026)	Controlled experimental study (University of Bonn, Fraunhofer IAIS; preprint).	EyePACS (35,000 unlabelled fundus images) for SSL; APTOS-19 (3,662 images, 5 DR grades) for fine-tuning and main evaluation.	SSL pretraining improved selective prediction over scratch training; pretraining length changed calibrated confidence and abstention quality, so it should be treated as a reliability design choice.	Retinal images, not MRI; no clinician deferral study.
Gutbrod et al. (2025)	Benchmark creation and evaluation of OOD detection in medical imaging (OTH Regensburg; CVPR 2025, with corrigendum).	Three medical imaging benchmarks across domains comprising 14 datasets split into covariate-shifted in-distribution, near-OOD and far-OOD sets.	Method rankings differed from natural-image benchmarks; near-OOD detection in medical images remained hard, and feature-based methods often did better than logit-based ones.	Detection only; not linked to deferral or clinical cost.
Xu et al. (2025)	Methodological and empirical study (New Jersey Institute of Technology, preprint).	Two public datasets (classification tasks) used to verify coverage and risk.	Both variants achieved target coverage and risk with nearly identical performance; SCRC-I was computationally more efficient; prediction sets on accepted cases were more compact.	No clinical evaluation; uniform costs.
Hekler et al. (2025)	Systematic empirical study of calibration across model generations (Goethe University Frankfurt, DKFZ; preprint).	ImageNet validation (90/10 test/calibration split), ImageNet-C (19 corruptions x 5 severities) and domain-specific biomedical imaging tasks with real distribution shifts.	Modern models were underconfident in-distribution (unlike the classic overconfidence) and better calibrated under shift; post-hoc calibration helped in-distribution but became less reliable and sometimes counterproductive under severe shift.	Classification only; no decision-level analysis.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Guarino et al. (2026)	Empirical study of uncertainty aggregation (MDC Berlin, DKFZ, Charité; preprint).	Diverse segmentation datasets (biomedical and driving, including OoD variants) with U-Net 3D, U-Net, HoVer-NeXt, HRNet and DeepLabv3+ models.	Spatially aware aggregators outperformed global averaging in both downstream tasks, but the best single aggregator depended on dataset; combining aggregators was more robust.	No link to clinical decisions or calibration.
Le et al. (2026)	Method development on glioma segmentation across datasets (University of Melbourne, Vietnamese universities; preprint).	BraTS and UT Southwestern glioma MRI datasets (cross-dataset shift).	Context sensitivity gave evidence of failure that uncertainty alone missed; combining both signals gave the strongest average failure detection, better than an ensemble.	Ranking only; no threshold or clinical evaluation.
Tailor et al. (2024)	Method paper with experiments on synthetic 2D data and three image classification benchmarks using simulated populations of experts.	Synthetic 2D clusters; Traffic Signs (GTSRB, train set down-sampled to 10,000); CIFAR-10; HAM10000 skin lesion images.	Reported in figures: L2D-Pop (both variants) gave higher expert accuracy on deferred examples and higher system accuracy than single-L2D on all three datasets, with the largest gap when experts' skills overlapped least.	Experts simulated from class-wise skills; real clinicians' changing behaviour not tested.
Mao et al. (2024a)	Theory paper (H-consistency bounds for multi-expert deferral surrogates) with experiments on two image benchmarks.	SVHN and CIFAR-10 image classification.	Cost type 1 - SVHN: 92.08 +/- 0.15, 93.18 +/- 0.18, 93.46 +/- 0.12 (1/2/3 experts); CIFAR-10: 73.31 +/- 0.21, 77.12 +/- 0.34, 78.71 +/- 0.43. Cost type 2 - SVHN: 92.36, 93.23, 93.36; CIFAR-10: 73.70, 76.29, 76.43.	Experts are other models, not humans; mainly theoretical; no workload limit.
Alves et al. (2024)	Method paper with simulation experiments on a large tabular fraud-detection dataset and synthetic expert teams.	Bank Account Fraud dataset (version 1): 1 million synthetic bank-account applications based on anonymised real data, 1.1% fraud prevalence, with concept drift over time.	Average 8.4% lower misclassification cost than baselines. Classifier h: ROC-AUC 0.70-0.76 and ECE 1.1-4.8% across scenarios. Expert-correctness models had similar ROC-AUC for HEM and OvA, but HEM needs far fewer human labels.	Fraud data and synthetic experts, not clinical; static capacity per batch.
Angelopoulos et al. (2024a)	Methodological / statistical paper with worked examples in vision and language.	Gut polyp segmentation pooled from Kvasir, Hyper-Kvasir, CVC-ColonDB, CVC-ClinicDB and ETIS-Larib (n = 1,000 calibration, 781 evaluation)	Polyp segmentation FNR 0.0987 +/- 0.0114 (target 0.1); MS COCO FNR 0.0996 +/- 0.0052; the realised risk tightly matched alpha with reasonable set sizes in all examples.	Guarantee is marginal (on average), not per patient; assumes exchangeable data.
Angelopoulos et al. (2025)	Statistical methodology paper with simulations and worked examples.	Synthetic autoregressive loss processes (V-shaped risk) for numerical comparison; MS COCO multi-label classification; MS COCO instance segmentation.	In simulations, multiple-testing LTT gave valid control with less conservativeness than uniform concentration; in worked examples, realised FDR, IoU-based risk and type-1 error stayed below targets with the stated probability.	Marginal guarantees; requires exchangeable calibration data and enough calibration size.
Borges et al. (2026)	Theory plus simulation and retrospective evaluation on six public medical segmentation tasks with in-distribution and OOD test sets.	Brain tumour: BraTS 2020 (3D nnU-Net); breast ultrasound: BUSI (UNeXt); skin lesions: ISIC 2018 (UNeXt); multiple sclerosis white-matter lesions: ISBI + MSSEG-1.	Simulation: SDC risk-coverage nearly identical to the optimal IDC. Real data: SDC was the best or near-best non-tunable estimator on all six tasks, robust on OOD sets, and competitive with tunable methods (values in Tables 3-5).	Binary or per-class Dice; image-level abstention only, not region-level escalation.
Lambert et al. (2024b)	Systematic search plus narrative / taxonomic review.	Not applicable (review of published studies on MRI, CT, PET, X-ray and other imaging).	Uncertainty quantification is an active field with many methods but no agreed evaluation standard; uncertainty arises from noisy images, imperfect labels, missing data and inter-site variability; integration into clinical workflow and user studies remain rare.	Review up to mid-2022; no quantitative meta-analysis.
Koutsoubis et al. (2025)	Narrative review.	Not applicable (review).	Identifies key gaps: privacy-performance trade-off, UQ under non-IID client data and class imbalance, need for calibrated outputs from FL models, and need for clinical validation of PPFL + UQ.	Lack of studies combining privacy-preserving FL with uncertainty quantification on real multi-site medical data.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Bangalore Yogananda et al. (2025)	Retrospective multi-institution development and external testing study.	2,481 pre-operative glioma cases with multi-contrast MRI and IDH status from 5 public sources (TCIA, EGD, UCSF, UPenn, IvyGAP) and in-house/collaborators (UTSW-1/2/3, NYU, UWM); 577 IDH-mutant and 1,904 wildtype.	TG-1 subject-wise AUC: NYU 0.9705, UWM 0.9599, EGD 0.9867, UPenn 0.9801, overall 0.9755; accuracies 96.4% and 95.1% on test groups; overall AUC 0.98; Brier score of confidence scores 0.0125.	IDH only (not full WHO 2021 integrated diagnosis); US/European cohorts; retrospective.
Elyassirad et al. (2025)	Retrospective development with internal and external validation.	UCSF glioma dataset (495 patients; WHO grade 2: 56, grade 3: 43, grade 4: 396) split 70/10/10/10 into train/validation/calibration/test.	UCSF test: NCST 0.8874, coverage 0.9982, UR-Dice correlation -0.827 ($p < 0.001$); uncertain 10 cases (20.4%) Dice 0.7090 vs certain 0.9023 ($p < 0.001$). UPenn external: Dice 0.8257, coverage 0.9977, significant negative correlation.	Two datasets; whole-tumour style segmentation; retrospective.
Wang et al. (2025a)	Retrospective single-institution development study (Duke University).	76 supratentorial meningioma patients treated with radiotherapy; GTVs contoured by one experienced radiation oncologist on high-resolution contrast-enhanced T1 MRI.	Sensitivity 0.758 vs 0.746; Dice 0.760 vs 0.742; mHD 0.612 vs 0.744 cm; HD95 2.682 vs 2.912 cm (SPU-Net vs U-Net); uncertainty maps highlighted ambiguous tumour boundaries.	Single centre, single annotator, 76 patients; meningioma rather than glioma.
Alshamlan (2026)	Retrospective evaluation on public datasets with a pre-documented analysis plan.	BRISC2025 brain MRI classification test partition (1,000 images); BraTS 2020 for segmentation (74 held-out cases) and glioma grading (high vs low grade).	Coverage 98.2%; selective accuracy 97.76% vs 96.60% full coverage; selective risk 3.40% -> 2.24%; withheld error rate 66.7%	Single operating point (continuous score AUROC not identifiable); 2D classification dataset; modules not evaluated together clinically.
Angelopoulos et al. (2024b)	Retrospective deployment simulation at a new hospital (preprint).	Head CT exams from Massachusetts General Hospital, scored by a head CT AI model developed previously by Do and colleagues (trained in a different environment).	False positives 233 (45%) -> 8 (5%); abstention (uncertain group) 14% of exams; guarantees held in a setting different from training.	Requires a labelled representative sample from the new site; preprint; single head CT task.
Gade et al. (2024)	Retrospective multi-centre study.	377 multi-centre 3T axial T2-weighted prostate MRI exams of biopsied men at risk of prostate cancer (age 66.64 +/- 7.47 years) with PI-RADS 2.1 ellipsoid prostate volume available.	RVD -2.81 +/- 8.85 (CP) vs -8.01 +/- 11.50 (DL), significant; mean difference 1.27 mL vs 6.07 mL; ICC 0.97 for PV_CP.	Prostate, not brain; volume assessment only; retrospective.
Montreuil et al. (2025)	Theoretical and empirical machine-learning study (NUS, Toulouse, ASTAR; ICLR 2026).	California Housing (two-stage regression), CIFAR-10, CIFAR-100 and SVHN (classification) with simulated experts and query costs.	Top-k and Top-k(x) consistently outperformed single-expert deferral, with better accuracy-cost trade-offs.	Simulated experts and non-medical data; no calibration of expert costs.
Alijani et al. (2026)	Methodological study with proof and two benchmarks (University of Victoria, preprint).	BraTS 2020 (multi-class: necrotic core, oedema, enhancing tumour); synthetic multi-organ CT benchmark built under covariate shift; latent-feature calibration/test sets of 66 and 73 cases for density-ratio estimation.	For every class on both benchmarks the 95% CI of observed coverage contained the nominal 90%, with narrower or more informative intervals than baselines.	Volumes only; small calibration sets; no clinical decision link.
Gaillochet et al. (2026)	Methodological study on public medical segmentation datasets (ÉTS Montréal, Polytechnique Montréal; preprint).	Multi-modal public medical segmentation datasets (details in paper).	Maintained marginal coverage while significantly improving the coherence and quality of prediction sets and stability with respect to the calibration parameter.	No brain-tumour-specific or clinical evaluation.
Mossina & Friedrich (2025)	Methodological study on a medical segmentation benchmark (IRT Saint Exupéry, preprint).	Polyps dataset collection with pretrained PraNet; each run splits test data in half (250 calibration images); 10 seeds.	Both conformal procedures achieved empirical validity close to the nominal 90% across tolerances, at the cost of shrinking masks.	Controls false positives only; 2D polyps.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Cheung et al. (2025)	Methodological study (Rice University, preprint).	Medical segmentation tasks where a downstream metric (e.g. organ or lesion size) matters; multiple public datasets.	Valid marginal coverage with tighter intervals than black-box conformal methods; recovered target coverage under covariate shift with learned weights.	No clinical decision evaluation.

Table 2.8: Evidence table for data-efficient, multi-source and federated models (O3): 30 representative studies.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Isensee et al. (2021) (base)	Method paper; automatic segmentation framework evaluated on the Medical Segmentation Decathlon (PDF read is the 2018 arXiv version; the 2021 Nature Methods version extends evaluation to 23 public datasets / 53 tasks).	Medical Segmentation Decathlon phase 1: BrainTumour (BraTS multimodal MRI, 3 labels: oedema, non-enhancing, enhancing), Heart, Liver, Hippocampus, Prostate, Lung, Pancreas; held-out test sets on the challenge platform.	Cross-validation Dice for BrainTumour labels 1/2/3: 2D U-Net 78.60 / 58.65 / 77.42; 3D U-Net 80.71 / 62.22 / 79.07. Heart 2D 91.36, 3D 92.45; Liver label 1 3D 94.11.	Needs strong GPU and no uncertainty output by default.
Baid et al. (2021) (base)	Challenge / benchmark description paper (RSNA-ASNR-MICCAI BraTS 2021): dataset creation, annotation protocol and evaluation framework for two tasks.	About 2,000 multi-institutional glioma cases (up from 660 in earlier BraTS) with pre-operative mpMRI: native T1, post-contrast T1 (T1Gd), T2, T2-FLAIR, from different scanners and protocols.	Not applicable for model results; provides the largest public annotated glioma mpMRI dataset (~2,000 cases) and a standard pipeline and metrics.	Mostly gliomas from Western centres, pre-processed and standardised.
Pati et al. (2022) (base)	Real-world global federated learning study across 71 institutions on 6 continents (FeTS initiative), with local validation and centralised out-of-sample evaluation.	25,256 MRI scans (>5 TB) of 6,314 glioblastoma patients from 71 sites (T1, T1Gd, T2, FLAIR); public initial model trained on 231 cases from 16 sites (BraTS)	Final consensus model improved over the public initial model by 33% for the surgically targetable tumour (tumour core) and 23% for the whole tumour extent on out-of-sample data; improvements also seen on local validation data at all sites (distributions in violin plots).	Large sites with good IT. Little on what happens with a handful of small sites.
Wang et al. (2023) (base)	Retrospective multicentre diagnostic study with training, validation, internal test and two external test cohorts.	2,624 patients: training 1,362 and validation 340 (FAHZZU), internal test 289 (FAHZZU), external test 1 305 (HPPH), external test 2 328 (XHCMU)	Task 1 AUCs: internal validation 0.953-0.995, internal test 0.932-0.994, external 1 0.923-0.987, external 2 0.904-0.952. Task 2 AUCs 0.938-0.983. Baselines ranged 0.793-0.997 on task 1.	Chinese cohorts only; no uncertainty estimation or deferral to pathologists.
Hindawi et al. (2025)	Real-world multi-centre feasibility study of federated learning (Roche with three hospital sites; Frontiers in Neurology 2025).	512 MRI cases from three sites for MS lesion segmentation, without sharing raw data.	Federated model achieved competitive segmentation; performance varied across sites due to heterogeneity; most loss reduction in first 10-15 rounds.	No calibration or uncertainty analysis; MS lesions rather than tumours.
Ren et al. (2025)	Retrospective federated multi-task study with external validation (Jiangnan University and partners; Frontiers in Radiology 2025).	Training on five public datasets (BraTS 2021 without overlaps, UPENN-GBM 671, TCGA-GBM, TCGA-LGG, REMBRANDT; n = 1,552) treated as separate institutions; external validation dataset of 466 patients.	Good segmentation and molecular-prediction performance on the external dataset of 466 patients.	Simulated institutions; no calibration analysis.
Shafique et al. (2025)	Method development and benchmarking against foundation models (Toronto Metropolitan University and partners; Scientific Reports 2025).	Training: 2,761 public breast cancer WSIs (CAMELYON16, CAMELYON17, HEROHE, TCGA-BRCA).	HistoLite had low representation shift between scanners and the lowest performance drop under scanner shift, with far less compute than foundation models, though large FMs had higher absolute accuracy in some settings.	Breast tissue; no uncertainty output.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Campanella et al. (2025)	Clinical benchmark of public pathology foundation models (Mount Sinai, MSKCC and partners; Nature Communications 2025).	Clinical slide datasets from routine practice across several institutions and cancer types, curated for diagnostic and biomarker tasks and released for benchmarking.	Foundation models improved over ImageNet features but no single model dominated all tasks; insights on best practices for training and selecting pretrained models; benchmark released.	No brain-specific tasks highlighted; no uncertainty.
Liu et al. (2026a)	Method development on BraTS 2021 (Nanjing University of Aeronautics and Astronautics; preprint).	BraTS 2021 multi-institutional multi-parametric glioma MRI (T1, T1ce, T2, FLAIR) with labelled fractions from 10% to 100%.	Validation DSC rose from 0.393 (10% labels) to 0.872 (100%); teacher 0.922; student surpassed teacher on subregions (NCR/NET 0.797, oedema 0.980) and recovered enhancing tumour (0.620) where the teacher failed.	Single dataset; uncertainty calibration not assessed.
Grande et al. (2026a)	Challenge method paper (PUCPR Brazil, University of Luxembourg; BraTS 2026 GoAT task, accepted to MICCAI 2026 BraTS cluster).	BraTS 2026 Generalizability Across Tumors (GoAT) data: heterogeneous brain tumour types and populations with labelled and unlabelled training cases; hidden validation via challenge platform.	Submitted configuration Dice/NSD 0.881/0.473 whole tumour and 0.817/0.490 tumour core (other regions in paper); self-training and augmentation each helped, best combined.	No calibrated uncertainty output; challenge data.
Grande et al. (2026b)	Controlled cross-dataset evaluation (PUCPR, Brazil; SIBGRAPI 2026).	Two well-known public brain tumour MRI datasets used as source and target for binary tumour vs non-tumour segmentation, stratified by tumour size.	FLAIR achieved the best cross-dataset performance (Dice >75%) across most tumour sizes; multi-sequence training improved further; even limited target-domain adaptation gave rapid gains.	Binary 2D segmentation; no uncertainty.
Zhang et al. (2025)	Dataset creation and domain-generalisation benchmark with a new method (Northwestern University; preprint).	PancreasDG: 563 3D MRI scans from six institutions with venous-phase and out-of-phase sequences; pixel-accurate pancreas masks from a double-blind two-pass protocol.	Cross-sequence shift posed fundamentally different challenges than cross-centre shift; limited sampling variance can be mistaken for shift; proposed method raised Dice from 43.55% to 70.39% and from 35.62% to 66.61% on two independent test centres.	Pancreas, not brain; no uncertainty.
Zhou et al. (2025)	Benchmark study of federated learning algorithms on medical image classification (Peking University Shenzhen; preprint).	Multiple medical imaging classification datasets (e.g.	Medical datasets posed substantial challenges; no single algorithm consistently performed best; the generative augmentation method improved robustness and efficiency.	Classification only; no calibration.
Verma et al. (2025)	Foundation model development and benchmarking (Icahn School of Medicine at Mount Sinai and collaborators; preprint).	Pretraining corpus of 585,657 H&E and Luxol fast blue-H&E WSIs; neuropathology data 23,144 slides (~800 million tiles) from Mount Sinai autopsy.	The neuropathology-specific model captured brain-tissue features better and outperformed general models on neuropathology tasks.	Mostly neurodegenerative tasks; tumour tasks less emphasised; no uncertainty.
Zenk et al. (2025)	International challenge with simulated FL (Task 1) and decentralised multi-site evaluation (Task 2).	Multi-institutional glioma mpMRI derived from BraTS with institution labels for partitioning (Task 1)	Task 1: teams using selective client sampling and adaptive aggregation consistently improved performance and training efficiency.	Glioma only; Task 1 FL is simulated; failure analysis per site is descriptive.
Shi & Wang (2025)	Method development on BraTS 2020 (Rensselaer Polytechnic Institute; preprint).	BraTS 2020 GBM: 309 scans from CBICA (103 train, 26 test) and TCIA (134 train, 33 test); public healthy brain images from multiple centres; 2D tumour slices every five slices; whole tumour in T2.	Synthetic shared data improved segmentation Dice across centres and supported fairness while protecting privacy.	2D; synthetic realism and residual privacy risk need clinical checking.
Rivera et al. (2026)	Benchmark creation with baseline experiments (University of Washington, Microsoft Health AI; preprint).	310 patients with multi-sequence brain MRI (T1, T1c, T2, FLAIR), of whom 95 have paired H&E WSIs and pathology reports; annotated with tumour type and grade and tumour masks; six clinically relevant glioma classes.	Multimodal fusion was feasible and modalities contributed complementary information across typing tasks.	Small paired subset; no calibration.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Linardos et al. (2025)	Challenge (benchmark) report, MICCAI 2024.	BraTS-derived multi-institutional glioma mpMRI: 1,251 training cases, 219 validation cases, 570 hidden test cases; labels ET, TC, WT; data partitioned by institution for FL.	Winner (PID-based): Dice ET 0.733, TC 0.761, WT 0.751; HD95 33.922 / 33.623 / 32.309 mm; convergence score 0.764 (best).	Simulated FL; absolute Dice lower than centralised models; glioma only.
Ma et al. (2025)	Foundation model development (fine-tuning SAM2) with benchmark evaluation and large human-in-the-loop annotation user studies.	Over 455,000 3D image-mask pairs (363,161 CT, 14,818 PET, 77,154 MRI) and 76,000 annotated video frames (19,232 ultrasound, 56,462 endoscopy).	MedSAM2 outperformed previous models across organs, lesions and modalities (SAM2.1 sizes showed no significant differences among themselves); human-in-the-loop annotation with MedSAM2 reduced manual effort by more than 85%.	Still needs a human prompt; not designed to give calibrated confidence.
Farahani et al. (2026)	Systematic review and diagnostic test accuracy meta-analysis.	1,517 unique publications screened from PubMed, Scopus, Embase, Web of Science and Google Scholar; 104 included, 72 in meta-analysis (many using TCGA/TCIA and institutional cohorts).	IDH: sensitivity 0.80 (95% CI 0.77-0.83), specificity 0.85 (0.81-0.87). 1p/19q: sensitivity 0.75 (0.65-0.82), specificity 0.82 (0.75-0.88). Significant moderators: tumour segmentation method and extent of DL integration.	Few studies with true external or prospective validation; heterogeneous MRI protocols.
Tampu et al. (2026)	Retrospective national multicentre evaluation of computational pathology pipelines.	1,449 H&E WSIs from 540 children and young adults (age 8.5 +/- 4.9 years; 284 male, 253 female) from six Swedish university hospitals (2013-2023), after excluding 62 cases / 128 WSIs; five scanners at 20x or 40x.	UNI + ABMIL MCC 0.76 / 0.63 / 0.60 for category / family / type; foundation features better than ResNet50 out-of-site; similar relative generalisation drop for all extractors.	Rare tumour types have few cases; scanner variability remains a source of error.
Saueressig et al. (2026)	Retrospective multimodal model development with independent paired test set.	WSI: EBRAINS (497 glioblastoma, 132 astrocytoma, 157 oligodendroglioma). MRI: UCSF-PDGM (398 / 84 / 15) and EGD (249 / 62 / 58).	MM-MoE AUC 0.98 (validation), 0.94 (test); random label-pairing equivalent to patient-matched training and better than naive ensembling; unpaired training equivalent except for late fusion; all models at least matched the 70% initial pathologist accuracy.	Only three WHO subtypes; TCGA is the only paired test cohort.
Piffer et al. (2026)	Retrospective single-centre development and evaluation study with clinical usability rating.	174 paediatric patients (0-18 years) at University Children's Hospital Zurich, 2014-2022, pre-treatment MRI (T1, T1-C, T2, FLAIR)	WT and T2H mean Dice 0.85 vs human 0.86; ET 0.75; CC 0.26; T1+T1-C+T2 close to full protocol; up to 83% time reduction.	Cystic components poorly segmented; single centre (Zurich).
LaBella et al. (2026)	Dataset descriptor for a public segmentation challenge.	750 RT planning MRI exams from 747 patients at Duke, UCSF, SUNY Upstate, University of Washington, University of Missouri.	Largest public meningioma RT planning MRI set: 570 public scans with 500 expert GTVs; heterogeneous acquisition parameters across and within sites.	Single sequence (T1c); GTV only, no clinical target volume or organs at risk.
Wu et al. (2025)	Method paper with multi-dataset evaluation.	17 medical segmentation tasks across CT, MRI, ultrasound, fundus and dermoscopy (e.g., BTCV multi-organ CT, BraTS brain tumour, REFUGE optic disc/cup, thyroid nodule ultrasound, ISIC melanoma).	Best or comparable average Dice across tasks with far fewer trainable parameters.	Still needs prompts and labelled fine-tuning data; compute of SAM backbone.
Zuo et al. (2025)	Retrospective single-centre study with public data (West China Hospital, Sichuan University; ethics No. 2021.1319, written consent).	472 H&E WSIs from 226 patients at West China Hospital (scanned on Hamamatsu NanoZoomer-SQ) and 1,604 WSIs from 880 TCGA patients; five subtypes (astrocytoma.	High AUCs for distinguishing grades of astrocytoma and oligodendroglioma and glioblastoma; IDH status inferred in TCGA with AUC 0.9488.	No external test of the local model; no uncertainty estimates.
Babendererde et al. (2026)	Real-world nationwide federated deployment across three German university hospitals (Cologne, Essen, Leipzig) with a new open-source framework (TU Darmstadt and partners; npj Digital Medicine 2026).	BCSS: 144 TCGA breast cancer H&E WSIs (0.25 um/px, 256x256 patches; per client 34 train, 7 validation, 7 test cases)	Global average Dice 0.764 (BCSS) and 0.754 (SemiCOL) despite heterogeneity and firewall constraints, close to centralised performance.	Segmentation of non-brain tissue; no calibration analysis after aggregation.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Touhami et al. (2026)	PRISMA systematic review (Multimedia University, KPJ Healthcare University, Malaysia); no registered protocol.	Studies (2020-2025) applying federated learning to histopathology image classification (e.g. LC25000, TCGA CRC-DX, BreakHis, HAM10000, TCGA, MC-CRC datasets).	FL achieves performance comparable to centralised learning in many studies, but issues include IID assumptions, non-IID performance drops, communication and encryption costs, and lack of large-scale validation; recommends standardised evaluation, efficient communication and real deployments.	No meta-analysis; few real deployments in included studies.
Bas & Ozturk-Isik (2026)	Retrospective comparison of centralised and federated learning (Bogazici University, Istanbul; Diagnostics 2026).	UCSF-PDGM: 501 pre-operative diffuse gliomas (103 IDH-mutant, 398 wild-type), 3T GE MRI (T1c IR-SPGR, 3D T2 FSE, FLAIR) with expert-corrected multi-compartment segmentations.	Context-preserving soft filtering substantially improved performance across training schemes; federated learning was a privacy-preserving alternative but with some performance cost relative to centralised training.	Simulated federation; 2D slices; no calibration analysis.
Karami et al. (2026)	Method development with benchmark and clinical MRI validation (Sharif University of Technology and partners; Scientific Reports 2026).	MNIST, CIFAR-10, an Alzheimer's MRI dataset (proof of concept) and the OASIS clinical brain-MRI cohort (clinical validation); simulated clients with IID, Dirichlet and label-skew (70%, 90%) partitions.	Detection F1 above 0.98 for gradient-scaling attacks while preserving accuracy under magnitude-preserving attacks where explicit detection was limited; robust under non-IID splits.	Classification, not tumour segmentation; adaptive parts not fully exercised in clinical runs.

Table 2.9: Evidence table for clinical evaluation of human-AI teams (O4): 30 representative studies.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Vasey et al. (2022) (base)	Consensus-based reporting guideline development (modified Delphi, consensus meeting, qualitative evaluation).	Expert panel from 20 pre-defined stakeholder categories: 123 experts in Delphi round 1, 138 in round 2, 16 at the virtual consensus meeting and 16 in the qualitative evaluation.	Final guideline with 17 AI-specific reporting items (28 subitems) and 10 generic items covering intended use, users and their training, AI system version and inputs/outputs, human factors (usability, learning curve, user trust), safety and errors, and human-AI agreement.	Guideline only, it does not say which metrics to use for deferral.
Chen et al. (2024b) (base)	Systematic review and meta-analysis.	36 included studies of image-based disease detection (e.g., mammography, chest radiography, other modalities) reporting reading time or reading quantity with and without AI.	Reading time -27.20%; reading quantity -44.47% (second reader) and -61.72% (pre-screening); relative sensitivity 1.12, specificity 1.00; substantial heterogeneity.	High heterogeneity and uneven study quality; few studies define reading time clearly; most are retrospective.
Keim-Malpass et al. (2026)	Pragmatic cluster-randomised controlled trial (University of Virginia, USA; NCT04359641), January 2021 to October 2022, reported per CONSORT-AI; IRB waiver of consent; DSMB oversight.	10,422 inpatient visits on an 85-bed acute cardiology and cardiac surgery ward; 11 clusters defined by room number.	No clear improvement in event-free hours overall; patients with large spikes in risk score had more events, and the pragmatic design allowed sicker patients to be moved towards displayed beds, complicating interpretation.	Passive display without linked actions; contamination from pragmatic bed allocation.
Alzyood (2026)	Short communication / conceptual framework (Oxford Brookes University, UK) in International Health.	Not applicable; synthesis of literature on AI adoption, digital health implementation and infection surveillance in resource-limited settings.	Provides policymakers and practitioners with a structured pathway for context-appropriate, human-centred AI implementation in LMICs.	Framework not empirically tested.
Onyeze et al. (2025)	Systematic review of diagnostic performance and implementation feasibility (Ninewells Hospital, Dundee, UK), PRISMA; studies January 2015 to June 2025 from six databases.	109 records, 86 screened, 58 full texts, 6 studies included from LMICs (World Bank FY25 classification), e.g.	High accuracy (e.g.	Only six studies; few RCTs or cost analyses.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Vijayan et al. (2026)	Narrative review of implementation experiences (Qure.ai Technologies, India and USA) in BJR|Open.	Published studies and programme reports on AI-assisted chest X-ray for TB case finding in LMICs.	AI CXR improved case detection, diagnostic accuracy and turnaround time in high-burden settings, identified subclinical TB, and enabled integrated workflows for multiple chest diseases and decentralised care.	No systematic method; vendor perspective.
Moehring et al. (2025)	Experimental data descriptor (MIT, Harvard, Stanford and partners) in Scientific Data; MIT COUHES exempt protocol E-2953.	Collab-CXR: 227 professional radiologists (mean 22 years' experience; 17% US-based; recruited including via a Vietnamese healthcare system) assessing subsets of 324 historical chest X-ray cases.	Largest known dataset comparing human-AI collaboration with AI or humans alone in radiology, with rich metadata on radiologist characteristics and decisions, released for research on how radiologists incorporate AI.	Remote experimental setting; single modality.
Chuang et al. (2026)	Retrospective multi-reader study comparing two AI assistance modes (Chang Gung Memorial Hospital and aetherAI, Taiwan); IRB 202000956B0.	70 nasopharyngeal biopsies (35 carcinoma, 35 benign) from three hospital branches, one H&E slide each, scanned at 40× (NanoZoomer S360); 60 cases used in the experiment; 10 board-certified pathologists.	AI-HPF mode kept high diagnostic accuracy and was robust to false-positive AI predictions, while assistance modes affected review time and perceived difficulty differently.	Binary task; no calibrated uncertainty display tested.
Riccardi et al. (2026)	Retrospective simulation study (conference abstract) in a population-based screening programme (Local Health Unit Roma 2 and Università Cattolica, Italy), funded by the Italian Ministry of Health.	38,075 screening mammograms from the 2025 cohort of the largest Italian Local Health Unit.	4,542 exams (11.9%) below threshold; 96% of these were negative at double reading, 201 recalled with 18 cancers; up to one extra cancer could be missed; workload reduction about 5% of annual reading volume.	Simulation only; threshold not optimised.
Patel et al. (2026)	Narrative review with structured search and dual screening (authors from USA, Pakistan and India) in Annals of Medicine & Surgery.	PubMed, Scopus and Google Scholar, January 2000 to May 2025, using connectome, neurosurgery, CNS tumour, brain mapping, tractography, fMRI and AI terms.	Connectome-guided neurosurgery shifts from lesion-centric to network-centric surgery, supporting function preservation; limitations include tractography errors, fMRI neurovascular uncoupling, cost and lack of standardisation.	Narrative; limited clinical outcome evidence.
Shi et al. (2026)	Nationwide cross-sectional survey of physicians in mainland China (Shuguang Hospital Shanghai, KINGYEE and Zhejiang University), January-April, distributed via Medlive.	4,024 in-service physicians from 29 provinces (77.4% tertiary hospitals; 72.9% senior titles).	Random forest accuracy 85.6% and AUC 0.836; SHAP identified organisational and expectancy factors (e.g. facilitating conditions, performance expectancy) as key drivers of acceptance; SEM confirmed direct and mediated effects.	Intentions, not actual use; senior-heavy sample.
Naicker et al. (2026)	Prospective qualitative implementation study across three phases (Queensland University of Technology and Princess Alexandra Hospital, Australia), reported per COREQ.	43 one-on-one interviews with 7 radiographers, 20 radiology registrars and 16 consultant radiologists (26 male, 17 female); 9 interviewed at multiple time points.	56 barriers/18 enablers at baseline, 55/14 peri-implementation and 82/33 post-implementation; clinicians stayed willing to engage, but sustainable adoption depended on consolidating early positive experiences and addressing negative ones such as workflow and performance issues.	Single site; commercial tool not named in detail.
Song et al. (2026)	Retrospective reader study of LLM assistance (Chonnam National University Hospitals, South Korea); IRB CNUHH-2024-176, consent waived.	127 histologically confirmed intra-axial brain tumour cases (2014-2023; mean age 40.3, range 1-87) covering common, moderate-incidence and rare subtypes; radiologists of different experience levels (trainees and experts).	Both LLMs had comparable diagnostic accuracy; LLM assistance was associated with improved trainee performance, with smaller effects for experienced radiologists, in this constrained setting.	Selected images, not full MRI volumes; no uncertainty handling.
Graham et al. (2026)	Retrospective non-interventional multi-centre multi-reader two-stage paired study (Histofy Ltd with UK, Lithuanian and Spanish hospitals); arXiv preprint.	WSIs from 385 unique patients at three independent sites, across multiple tumour types; multiple pathologists, each slide read by three randomly selected readers in both stages.	AI assistance improved reproducibility and reduced assessment time across tumour types; the frequency of score change between unassisted and assisted assessment was comparable with routine inter-pathologist variation.	Ground truth by consensus not independent stain.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Wu et al. (2026a)	Multicentre prospective randomised three-arm parallel-group multireader study at three hospitals in China (Nanchang University and HKUST), July-September 2026; ethics 2026-756, ChiCTR2600129243; arXiv preprint.	154 residents assessed, 132 randomised (1:1:1 after specialty stratification), 123 analysed (60 radiology, 63 non-radiology; < 3 years' experience; mean age 24.1).	Dual-suggestion support effects differed by specialty (interaction 10.44 percentage points, 95% CI 4.36-16.52, $p < 0.001$), with benefits particularly when the shared GPT suggestion was incorrect, helping some residents avoid following a wrong single suggestion.	Residents only; no expert comparison; recruitment-based sample size.
Raees & Papangelis (2026)	Systematic literature review following PRISMA (Rochester Institute of Technology, USA); arXiv preprint.	729 search results from Scopus (mid-2025) and ACM Digital Library (December 2025), 2018 onwards; 22 empirical studies of appropriate reliance included.	Trust measures do not capture appropriate reliance; reliance constructs are fragmented; most studies (20 of 22) used lay users or crowd workers; authors argue for consensus metrics to compare studies.	Few studies with domain experts such as clinicians.
Rosbach et al. (2026)	Web-based 2×2 within-subjects experiment (TH Ingolstadt, Vetmeduni Vienna, FU Berlin, Flensburg UAS); Machine Learning for Biomedical Imaging; arXiv preprint.	28 trained pathology experts estimating tumour cell percentage (0-100%) on H&E tissue patches.	AI improved overall accuracy but induced automation and anchoring bias; time pressure increased severity of reliance; LMMs showed professional experience and self-efficacy were associated with automation reliance.	Simulated AI; single estimation task.
Nicolson et al. (2025)	Three-stage within-subject reader study.	Fetal ultrasound images for gestational age estimation; 10 sonographers.	MAE 23.5 days alone, 15.7 with prediction, 14.3 with explanation (non-significant vs prediction only); heterogeneous individual effects; higher confidence but no significant change in trust or reliance.	Only 10 participants; single task (gestational age); immediate effects only.
Brändl et al. (2025)	Method development with retrospective validation and a small prospective intraoperative series.	Heidelberg brain tumour classifier v11b4 reference methylation set (91 CNS classes) for training; over 200 brain tumour samples nanopore-sequenced for validation.	94.52% accuracy (91 classes) and 97.72% (grouped classes) at 7,500 CpGs; robust to 1-10% calling error; results within 15 min of sequencing; better than Sturgeon and nanoDx in very short runs; errors mainly within closely related classes.	Prospective evidence still limited (10 intraoperative cases); needs sequencing infrastructure.
Sounderajah et al. (2025)	Reporting guideline development (modified Delphi consensus).	Delphi round 1: 209 participants (45 items); round 2: 143 participants (26 items); over 240 international stakeholders overall.	40 essential items; 14 new and 4 modified AI-specific items.	Reporting standard only; does not judge study quality.
Kwon et al. (2026)	Single-centre retrospective multi-reader multi-case study (Seoul National University Bundang Hospital; IRB E-2310-858-301, consent waived).	258 diagnostic breast ultrasound examinations (2015-2020): 129 malignant (112 invasive, 17 DCIS, pathology-proven) and 129 benign (55 biopsy-proven, 74 stable >=12 months); index lesion per case.	Modest AUROC improvement and shorter interpretation time with AI; accuracy, sensitivity and specificity did not change significantly.	No confidence-based deferral; balanced enriched set.
Kunz et al. (2025)	Prospective non-clinical user experience study in a simulated ICU setting (Fresenius Medical Care with Marienhospital Bottrop, Germany).	8 healthcare professionals evaluating a CKRT clinical decision support prototype in simulated scenarios.	Mean SUS 87.5 (SD 8.39, 'excellent'); positive UEQ scores for attractiveness and task-related dimensions; trend to lower total workload mainly from less physical demand and effort.	Very small sample; industry-led; not AI with uncertainty.
Al-Anezi (2026)	Systematic review following PRISMA 2020 and Tranfield et al. principles (Imam Abdulrahman Bin Faisal University, Saudi Arabia).	11,269 records from PubMed, Scopus, Web of Science, ScienceDirect, IEEE Xplore and Google Scholar (January 2015 to June 2025); 29 studies (2022-2025) included.	GenAI improves efficiency and decision support but brings automation bias and risk of eroding diagnostic expertise; mitigation includes human-in-the-loop frameworks, explainable AI, ethical governance and continuous training.	Little empirical longitudinal evidence on deskilling; no quantitative pooling.
Kim et al. (2025a)	Retrospective reader study with crossover design (Technical University of Munich, Germany); IRB approved, consent waived.	20 3D TOF-MRA studies (06/2021-12/2023) from an outpatient practice using mdbrain v4 routinely: 10 cases with at least one false-positive AI finding.	False-positive AI findings significantly raised suspicion ratings ($p = 0.01$); inexperienced readers rating such cases unremarkable fell from 63.3% to 23.3%; reading times shorter with AI (inexperienced 164.1 vs 228.2 s, $p < 0.001$).	No uncertainty or explanation display tested as mitigation.

Study	Design and setting	Data and sample	Key quantitative results	Limitation for this study
Davis & Salwei (2026)	Perspective / conceptual paper (Vanderbilt University Medical Center, USA) in JAMIA.	Not applicable (conceptual framework drawing on literature).	Argues that explanation of context, rather than only of features, helps clinicians calibrate appropriate trust and reliance, avoiding both automation bias and under-reliance, without adding cognitive burden.	Framework not yet empirically evaluated.
Cicin & Çetin Gürkan (2026)	Cross-sectional survey with structural equation modelling (Istinye University, Turkey), integrating the Theory of Planned Behaviour, Technology Acceptance Model and trust theory.	414 physicians from 38 cities in Turkey.	Trust fully mediated the effect of perceived behavioural control ($p < 0.001$) and other constructs on intention; subjective norms and perceived behavioural control had weaker direct effects ($p < 0.05$ and $p = 0.07$).	Intention, not actual use; single country.
Lin-Martore et al. (2025)	Qualitative interview study with interpretivist approach and template analysis (University of California, San Francisco, USA), November 2023 to April 2024.	14 academic paediatric emergency medicine physicians (1-34 years' experience) using POCUS across three urban paediatric EDs.	Five factor themes shaped trust: technological (accuracy, transparency), contextual, user, social and organisational; perceived benefits included image acquisition guidance, landmark identification, interpretation, measurements and reduced mental load.	Hypothetical AI; single institution.
Schulz et al. (2025)	Cross-sectional online survey (Nanyang Technological University, Singapore, with Asia-Pacific partners); IRB approved (IRB-2022-756).	319 gastroenterologists and GI surgeons trained in colonoscopy from ten Asia-Pacific countries/regions.	Senior clinicians with low risk perception were highly accepting of AI, whereas seniors with high risk perception were substantially less accepting; acceptance fell as AI tasks became more sophisticated.	Intentions only; scenarios hypothetical.
Boaro et al. (2025)	Cross-sectional global online survey (University of Verona, Leiden University and EANS ETIN Task Force), reported per CHERRIES.	248 complete responses from neurosurgeons and trainees in 61 countries (69% Europe; 62% post-residency; 68% academic hospitals).	Knowledge of basic AI concepts was limited, but interest and attitude toward AI implementation were strongly positive; respondents called for systematic AI training and guidelines.	Few respondents from low- and middle-income countries.
Chan et al. (2025)	Systematic review following PRISMA (Stanford Neurosurgical AI and ML Laboratory, USA).	PubMed search on 8 February 2025 (publications from 2016) returning 267 articles; 86 studies releasing novel public or access-on-application neurosurgical datasets included.	86 public datasets catalogued, dominated by neuro-oncology imaging; baseline models from these datasets are limited by poor external validation, lack of reproducibility and reliance on suboptimal evaluation metrics; gaps in several neurosurgical subspecialties and populations.	PubMed only; underrepresentation of LMIC data.

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